

PRACTITIONER KNOWLEDGE REPOSITORY
on Banking Regulation and Risk Management

MODULE D

Practitioner's Dictionary

XVA, Banking Regulation & Risk Management

Layer 1	Layer 2	Layer 3
Plain English — what it is in plain language	Formula or Structure — the quantitative definition	Practitioner Usage — how it works in the real world
Layer 4		
Interview Answer — how to answer it confidently in 45 seconds		

NOTE OF CAUTION

This dictionary is designed as a creative learning aid for newcomers to derivatives, XVA, and banking regulation. Definitions, formulae, and interview answers are intended to build intuition and understanding — they are not legal, accounting, or regulatory advice.

Practitioners must always refer to current ISDA documentation, Basel Committee publications, and applicable regulatory guidance for authoritative definitions. The 20-60-20 methodology applies: all content has undergone human review but may not reflect the most current regulatory or market practice at the time of reading. When in doubt — check the primary source.

Curated by Sarthaka Nanda (with Anthropic Claude)
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Alphabetical Index

This dictionary contains 87 entries across XVA & valuation adjustments, Basel capital framework, market risk, counterparty credit risk, collateral and legal documentation, credit risk limits, and risk monitoring infrastructure. Four-layer entries with interview answers throughout.

Abbreviation	Long Form	Module	Illustrated
AAD	Adjoint Algorithmic Differentiation	M6	—
Add-on (PFE)	Potential Future Exposure Add-on — SA-CCR Component	M3	—
Aggregate Limit	Aggregate Credit Limit	Risk	—
Backtesting	Model Backtesting	M2	—
Basel IV / Endgame	Basel IV — Final Basel III Reforms / US Basel Endgame	M1,M6	—
Basis Risk	Basis Risk	M3,M6	—
bCVA	bCVA — Bilateral Credit Valuation Adjustment	M3	—
Breach	Limit Breach / Limit Excession	Risk	—
Capital Greeks	Capital Greeks — KVA Sensitivities	M0,M6	—
CCB	Capital Conservation Buffer	M1,M6	—
CCP	Central Counterparty Clearing House	M3,M5	✓
CEM	CEM — Current Exposure Method	M3	—
CET1	Common Equity Tier 1	M1,M6	—
Cherry-Picking	Cherry-Picking — Selective Enforcement of Derivative Contracts	M5	—
Clean Risk	Clean Risk — Current Mark-to-Market Exposure	Risk	—
Close-out Netting	Close-out Netting — ISDA Section 6	M5	✓
CoVA	Collateral Valuation Adjustment	M6	—
Compression	Portfolio Compression — Notional Reduction	M6,Risk	—
Concentration Risk	Concentration Risk	Risk	—
CS01	CS01 — Credit Spread 01 (1 Basis Point Credit Sensitivity)	M3	—
CVA	Credit Valuation Adjustment	M3	—
Default Fund	CCP Default Fund	M3	—
Delta	Delta — First-Order Price Sensitivity	M0	—

Dirty Risk	Dirty Risk — Total Credit Exposure Including Settlement	Risk	—
Dodd-Frank	Dodd-Frank Wall Street Reform and Consumer Protection Act	M5	—
DV01	DV01 — Dollar Value of 01 (Interest Rate Sensitivity)	M0	—
DVA	Debit Valuation Adjustment	M3	—
EAD	EAD — Exposure at Default	M3	—
EMIR	EMIR — European Market Infrastructure Regulation	M5	—
ENE	ENE — Expected Negative Exposure	M3	—
EPE	EPE — Expected Positive Exposure	M3	✓
ES	ES — Expected Shortfall (Conditional VaR)	M2	—
FCA/FBA	FCA — Funding Cost Adjustment / FBA — Funding Benefit Adjustment	M4	—
FRTB	FRTB — Fundamental Review of the Trading Book	M2	—
FVA	FVA — Funding Valuation Adjustment	M4	—
G-SIB	G-SIB — Global Systemically Important Bank	M1,M6	—
Gamma	Gamma — Second-Order Price Sensitivity	M0	—
Greeks	Greeks — Derivative Risk Sensitivities	M0	—
Haircut	Haircut — Collateral Valuation Discount	M5	—
Headroom	Headroom — Available Limit Capacity	Risk	—
Hurdle Rate	Hurdle Rate — Required Return on Equity	M6	—
IM	IM — Initial Margin	M4,M5	—
IM Amortisation	IM Amortisation — Margin Decay Over Trade Life	M4,M5	✓
IMM	IMM — Internal Model Method	M1,M3	—
IRB	IRB — Internal Ratings-Based Approach	M1	—
ISDA Master	ISDA Master Agreement	M5	✓
Issuer Risk	Issuer Risk — Credit Risk from Bond/Note Holdings	Risk	—
KVA	KVA — Capital Valuation Adjustment (Kapital)	M6	—
Leverage Ratio	Leverage Ratio — Non-Risk-Sensitive Capital Backstop	M1,M6	—
LGD	LGD — Loss Given Default	M3	—
Limits Framework	Limits Framework — Credit and Market Risk Limit Hierarchy	Risk	✓

MDDR	MDDR — Maximum Daily Deal Rate	Risk	—
MM Limit	MM Limit — Mark-to-Market Exposure Limit	Risk	—
MPoR	MPoR — Margin Period of Risk	M4,M5	—
MTA	MTA — Minimum Transfer Amount	M4,M5	—
MtM	MtM — Mark-to-Market	M3,M5	—
MVA	MVA — Margin Valuation Adjustment	M4	—
Netting Opinion	Netting Opinion — Legal Enforceability Confirmation	M5	—
Netting Set	Netting Set — SA-CCR Unit of Calculation	M5	—
OBSI	OBSI — Outstanding Business and Settlement Indicator	Risk	✓
OIS	OIS — Overnight Index Swap	M4,M5	—
Output Floor	Output Floor — Basel IV RWA Minimum	M1,M6	—
P&L Attribution	P&L Attribution — XVA Profit and Loss Decomposition	M6	—
PD	PD — Probability of Default	M3	—
PFE	PFE — Potential Future Exposure	M3,Risk	—
RC	RC — Replacement Cost (SA-CCR)	M3	—
Rehypothecation	Rehypothecation — Collateral Reuse	M5	—
Rho	Rho — Interest Rate Sensitivity	M0	—
RWA	RWA — Risk-Weighted Assets	M1,M6	—
RWR	RWR — Right-Way Risk	M3	—
SA-CCR	SA-CCR — Standardised Approach for Counterparty Credit Risk	M3	—
SCB	SCB — Stress Capital Buffer	M1,M6	—
Settlement Risk	Settlement Risk — Herstatt Risk	Risk	—
SFTR	SFTR — Securities Financing Transactions Regulation	M5	—
SIMM	SIMM — Standard Initial Margin Model	M4	—
SIMM Recalibration	SIMM Recalibration — Semi-Annual Risk Weight Updates	M4	—
Tenor Limit	Tenor Limit — Maximum Maturity Limit	Risk	—
Theta	Theta — Time Decay	M0	—
Threshold (CSA)	Threshold — CSA Credit Exposure Threshold	M4,M5	—
UMR	UMR — Uncleared Margin Rules	M4,M5	—

Utilisation	Utilisation — Limit Utilisation Percentage	Risk	—
VaR	VaR — Value at Risk	M2	—
Vega	Vega — Volatility Sensitivity	M0	—
VM	VM — Variation Margin	M4,M5	—
WWR	WWR — Wrong-Way Risk	M3	—
XVA	XVA — X-Valuation Adjustment	M3,M4,M5,M6	✓
XVA Desk	XVA Desk — Centralised Valuation Adjustment Function	M6	—

Dictionary Entries — A to Z

A

AAD — Adjoint Algorithmic Differentiation [M6]

Layer 1 — Plain English

A mathematical technique that computes the derivative of a function with respect to all its inputs simultaneously in a single backward pass through the computation graph — at approximately 2–5x the cost of one forward evaluation, regardless of how many risk factors there are.

Layer 2 — Formula / Structure

Cost ratio: AAD $\sim O(1)$ vs Finite Difference $\sim O(N)$ simulations for N risk factors. Speed-up $\sim N/\text{constant} \sim 100\times$ for $N=500$

Layer 3 — Practitioner Usage

AAD is the industry standard for XVA Greeks. Without it, computing $dKVA/dRate$, $dCVA/dSpread$, or $dColVA/dBasis$ for a large book requires $N+1$ full Monte Carlo simulations — computationally intractable intraday. AAD delivers all N sensitivities in one backward sweep. Standard reference: Savine (2018). Key vendors: ComptibL, Quaternion/MSCI, Numerix. ColVA Greeks share the same AAD sweep as KVA Greeks at negligible marginal cost.

INTERVIEW ANSWER

"AAD is a computational technique that calculates all risk sensitivities — all Greeks — in one backward pass through the simulation, at roughly 3-5x the cost of a single forward run rather than $N+1$ runs. For an XVA book with 500 risk factors, that's a 100x speed-up. Without AAD, real-time or intraday KVA Greeks are simply not feasible. The key idea is that the forward simulation records a computational tape of every operation; the adjoint backward sweep propagates derivatives from the output back to every input simultaneously."

Add-on (PFE) — Potential Future Exposure Add-on — SA-CCR Component ^[M3]

Layer 1 — Plain English

The component of SA-CCR EAD that captures the risk of future market moves increasing the exposure above the current replacement cost. The add-on is asset-class specific and reflects supervisory estimates of future volatility.

Layer 2 — Formula / Structure

Add-on = Effective Notional × Supervisory Delta × Maturity Factor × Supervisory Factor. EAD = 1.4 × (RC + PFE) where PFE = multiplier × Aggregate Add-on

Layer 3 — Practitioner Usage

Practitioners distinguish RC (current exposure — what you'd lose today) from Add-on (future potential exposure growth). The add-on increases with notional, tenor, and volatility. SA-CCR uses prescribed supervisory factors by asset class — e.g. 0.50% for IR derivatives — rather than bank-specific models. The alpha multiplier of 1.4 adds a further conservative buffer.

INTERVIEW ANSWER

"The SA-CCR add-on is the potential future exposure component of EAD — it estimates how much the trade's value might move against you between now and default. It uses a prescribed supervisory factor, the effective notional, and a maturity adjustment. Combined with the replacement cost and multiplied by alpha of 1.4, you get EAD. The key point for interviews: SA-CCR replaced CEM precisely because CEM's add-ons didn't recognise netting benefit in the PFE component — SA-CCR does."

Aggregate Limit — Aggregate Credit Limit ^[Risk]

Layer 1 — Plain English

A pre-approved maximum total credit exposure to a single counterparty across all products, tenors, and transaction types. When the aggregate limit is reached, no new transactions can be executed regardless of individual transaction size.

Layer 3 — Practitioner Usage

Aggregate limits are set by the credit committee and reviewed at least annually. They sit above product-level limits (OBSI, MM, MDDR) as the ultimate backstop. Headroom = Limit – current utilisation. Breach triggers immediate escalation and typically freezes new business.

INTERVIEW ANSWER

"An aggregate limit is the total credit exposure a bank is prepared to accept to a single counterparty across everything — derivatives, bonds, securities financing. It's the hard ceiling. Within that ceiling there are sub-limits by product type or tenor. The key concept: a limit breach doesn't mean you've lost money — it means the exposure has reached the level the credit committee pre-approved as the maximum acceptable risk."

B

Backtesting — Model Backtesting [M2]

Layer 1 — Plain English

The process of comparing a risk model's predictions against actual outcomes to verify model accuracy. For VaR models, backtesting counts the number of days actual losses exceeded the VaR estimate — too many exceptions triggers regulatory capital penalties.

Layer 2 — Formula / Structure

VaR exception = day when actual loss > VaR estimate. Basel traffic light: 0-4 exceptions (green zone), 5-9 (amber — capital multiplier uplift), 10+ (red — IMA approval at risk)

Layer 3 — Practitioner Usage

Backtesting is mandatory for FRTB IMA approval and must cover at least 250 trading days. A model failing backtesting loses desk-level IMA approval and reverts to the standardised approach, typically increasing capital significantly. Under FRTB, the P&L attribution test (comparing risk-theoretical P&L vs actual P&L) is equally important and harder to pass than backtesting alone.

INTERVIEW ANSWER

"Backtesting checks whether a VaR model is accurate by counting how often actual daily losses exceeded the predicted VaR threshold. Basel's traffic light framework allows up to 4 exceptions

in 250 days in the green zone — more than that and your capital multiplier increases. Under FRTB, backtesting is done at desk level not firm level, and you also need to pass the P&L attribution test. The key message: backtesting is not about catching every loss — it's about verifying the model's statistical properties."

Basel IV / Endgame — Basel IV — Final Basel III Reforms / US Basel Endgame ^[M1,M6]

Layer 1 — Plain English

The final tranche of Basel III reforms finalised by the BCBS in 2017 and being phased in from 2025 to 2030. Key provisions: the output floor (RWA cannot be less than 72.5% of SA), revised credit risk SA, revised operational risk framework, and revised CVA framework. In the US, the "Basel Endgame" proposal by the Federal Reserve implements these reforms with additional US-specific requirements.

Layer 2 — Formula / Structure

Output floor: Effective RWA = max(Internal Model RWA, 72.5% × SA RWA). Phase-in: 50% floor 2025 → 72.5% floor 2030.

Layer 3 — Practitioner Usage

Basel IV is the single largest driver of regulatory capital change for derivative-heavy banks since Basel III itself. The output floor constrains IMM benefit for CCR and IRB benefit for credit risk — both previously key sources of RWA optimisation. For XVA desks, Basel IV increases effective RWA and therefore KVA across the entire book. The US Endgame proposal initially proposed more aggressive implementation; after industry pushback a revised proposal was expected in 2025.

INTERVIEW ANSWER

"Basel IV — sometimes called the Basel Endgame in the US — is the final round of post-crisis reforms. The headline provision is the output floor: internal model RWA can't be less than 72.5% of standardised RWA. That constrains the benefit of IMM and IRB models that banks have used to reduce capital. For derivatives: if your IMM produces lower RWA than SA-CCR, the floor kicks in and you pay SA-CCR-equivalent capital anyway. The strategic response is to push more business through CCPs where the capital treatment is better even on a standardised basis, and to invest in SA-CCR optimisation rather than model sophistication."

Basis Risk — Basis Risk [M3,M6]

Layer 1 — Plain English

The residual risk that remains when a hedge does not perfectly offset the exposure being hedged because the two instruments move differently. Even when the direction of movement is the same, differences in magnitude, timing, or credit quality create basis risk.

Layer 3 — Practitioner Usage

Basis risk is endemic in XVA hedging. CVA desks hedge single-name credit risk using CDS indices — the index/single-name basis creates unhedged residual exposure. ColVA is driven by cross-currency basis risk — the spread between EUR OIS and USD OIS (converted through the cross-currency basis swap). The 2022 UK gilts crisis demonstrated that even "risk-free" basis (gilt vs swap) can be violently non-zero under stress.

INTERVIEW ANSWER

"Basis risk is the risk that your hedge doesn't move exactly in line with your exposure. It's unavoidable in practice because you rarely have a perfect offsetting instrument. The classic XVA example: hedging single-name CVA with an index CDS. The index moves when average credit deteriorates, but your specific counterparty might move differently. That gap is basis risk. In interviews, demonstrating awareness of basis risk in your hedging strategy signals practical understanding beyond theory."

bCVA — bCVA — Bilateral Credit Valuation Adjustment [M3]

Layer 1 — Plain English

The combined CVA and DVA adjustment that captures both the cost of counterparty default (CVA) and the benefit of one's own potential default (DVA). $bCVA = CVA - DVA$. For a symmetric relationship bCVA reflects the net bilateral credit risk embedded in the derivative.

Layer 2 — Formula / Structure

$$bCVA = CVA - DVA = -\int LGD_{cpty} \times PD_{cpty}(t) \times EPE(t) \times df(t) dt + \int LGD_{own} \times PD_{own}(t) \times ENE(t) \times df(t) dt$$

Layer 3 — Practitioner Usage

bCVA was the conceptually "complete" bilateral adjustment before FVA entered the framework. In a perfectly symmetric world where both parties use each other's credit risk, $bCVA + FVA = \text{total bilateral adjustment}$. In practice, DVA's controversial accounting treatment means bCVA is often decomposed back into CVA (booked, hedged) and DVA (booked, largely unhedged). FRTB CVA capital covers CVA risk — the risk that CVA marks change — not DVA risk.

INTERVIEW ANSWER

"bCVA is simply CVA minus DVA — the bilateral adjustment that captures counterparty default risk from both sides. CVA is the expected cost if the counterparty defaults; DVA is the expected benefit if you default. Together they give the bilateral view. The controversy: DVA creates profit when your own credit deteriorates, which feels perverse. That's why FRTB CVA capital only covers CVA risk, not DVA — regulators don't want banks holding capital for the benefit of their own potential default. In interviews: distinguish bilateral CVA (economic concept) from the FRTB CVA capital charge (regulatory concept) — they're related but not the same thing."

Breach — Limit Breach / Limit Excession [Risk]

Layer 1 — Plain English

An excession of a credit or market risk limit — exposure utilisation exceeds 100% of the approved limit. Breaches must be immediately escalated to the credit officer and senior management.

Layer 3 — Practitioner Usage

Breach severity protocols: soft breach (100–110%) may allow new business subject to credit officer approval; hard breach (>110%) typically freezes all new transactions until resolved. Resolution options: trade reduction, collateral posting, or limit increase request with full credit analysis. Breach history is a key input to annual credit reviews.

INTERVIEW ANSWER

"A limit breach occurs when exposure exceeds the pre-approved credit limit. The critical thing interviewers want to hear is the process: identify the breach immediately, escalate to the credit officer, document the cause, determine whether new business is frozen or conditionally permitted, and establish a remediation plan. Breaches are not failures of risk management — they're the system working correctly. The failure would be ignoring them."



Capital Greeks — Capital Greeks — KVA Sensitivities [M0,M6]

Layer 1 — Plain English

The sensitivities of KVA to changes in underlying market parameters — analogous to the standard option Greeks but applied to the regulatory capital cost of a derivative. Capital Greeks quantify how KVA changes as rates, credit spreads, volatility, or notional change.

Layer 2 — Formula / Structure

$dKVA/dRate$ (IR Delta): KVA sensitivity to yield curve shift
 $dKVA/dCS01$ (Credit Sensitivity): KVA sensitivity to counterparty spread
 $dKVA/dVol$ (Vega): KVA sensitivity to implied volatility
 $dKVA/dNotional$ (Attribution): marginal KVA per unit of notional

Layer 3 — Practitioner Usage

Capital Greeks are computationally expensive under finite difference (each requires a full Monte Carlo re-simulation) but tractable under AAD (all delivered in one backward sweep). $dKVA/dRate$ is needed to hedge the interest rate sensitivity of the capital profile. $dKVA/dCS01$ shows how KVA changes as a counterparty deteriorates. $dKVA/dNotional$ enables trade-level KVA attribution — essential for pricing new trades at the margin. $dKVA/dVol$ is relevant for swaption and exotic books where SA-CCR EAD is volatility-sensitive.

INTERVIEW ANSWER

"Capital Greeks are the sensitivities of KVA to market risk factors — the same concept as standard option Greeks but applied to the regulatory capital cost. $dKVA/dRate$ tells you how much KVA changes per basis point move in rates — you need this to hedge the capital profile's interest rate sensitivity. $dKVA/dCS01$ shows KVA sensitivity to counterparty credit spread. $dKVA/dVol$ matters for options books where PFE is vol-sensitive. $dKVA/dNotional$ gives trade-level attribution — the marginal KVA of each new trade. The key operational point: AAD computes all of these simultaneously in one backward pass. Without AAD, each Greek requires a full re-simulation — completely intractable for intraday production."

CCB — Capital Conservation Buffer [M1,M6]

Layer 1 — Plain English

A 2.5% CET1 capital buffer required above the 4.5% minimum under Basel III, bringing the base minimum CET1 to 7%. Banks that draw on this buffer face automatic restrictions on dividends and bonus payments until it is restored.

Layer 2 — Formula / Structure

Total minimum CET1 = 4.5% (minimum) + 2.5% (CCB) = 7.0% (before G-SIB surcharge or SCB)

Layer 3 — Practitioner Usage

The CCB was designed to ensure banks accumulate capital in good times. For KVA, the CCB is part of CET1_req: $K(t) = RWA(t) \times CET1_req$. A G-SIB Bucket 2 bank has $CET1_req = 4.5\% + 2.5\% + 1.5\% = 8.5\%$ — every trade's capital cost is calculated against this full stack.

INTERVIEW ANSWER

"The Capital Conservation Buffer is an additional 2.5% CET1 on top of the 4.5% minimum — bringing the base to 7%. Its purpose is countercyclical: build it in good times so banks can absorb losses in stress without breaching the hard minimum. The mechanism is automatic restrictions on distributions if the buffer is drawn down. For interviews at a derivatives bank, the practical point is that the CCB raises the all-in capital requirement and therefore increases KVA on every trade."

CCP — Central Counterparty Clearing House [M3,M5]**Layer 1 — Plain English**

A financial market infrastructure that interposes itself between buyer and seller in a financial transaction, becoming the buyer to every seller and seller to every buyer. CCPs eliminate bilateral counterparty credit risk by mutualising it through a structured loss waterfall.

◎ CCP Loss Waterfall

1st line: Defaulting Member VM (daily MtM settlement)
 2nd line: Defaulting Member IM (MPoR buffer)
 3rd line: Defaulting Member Default Fund contribution
 4th line: CCP Equity — "skin in the game"
 5th line: Surviving Members Default Fund (pro-rata haircutting)

Each layer must be exhausted before the next is touched.
 CCP equity between defaulter and survivors = key design feature.

Layer 2 — Formula / Structure

CCP loss waterfall: Defaulter VM → Defaulter IM → Defaulter Default Fund → CCP Equity (skin in the game) → Surviving Members Default Fund (haircutting)

Layer 3 — Practitioner Usage

Post-2008 G20 mandates require standardised OTC derivatives to be centrally cleared. Cleared trades benefit from: reduced SA-CCR EAD (5-day MPoR vs 10-day bilateral), FRTB CVA exemption, and near-zero KVA. The concentration of systemic risk in CCPs themselves — too-big-to-fail risk shifting to FMIs — is the subject of ongoing regulatory debate.

INTERVIEW ANSWER

"A CCP sits in the middle of every cleared trade — it's the buyer to every seller and seller to every buyer. The key benefits: bilateral CCR is eliminated, replaced by CCP exposure which carries much lower capital treatment. The loss waterfall goes VM, then IM, then default fund. For the interview: the regulatory driver post-2008 was the G20 Pittsburgh commitment to move standardised OTC derivatives to CCPs to reduce systemic risk. The question to raise is whether concentrating risk in CCPs creates a new systemic risk rather than eliminating the old one."

CEM — CEM — Current Exposure Method [M3]

Layer 1 — Plain English

The predecessor to SA-CCR for measuring counterparty credit risk EAD. CEM calculated EAD as the sum of current MtM (floored at zero) plus an add-on based on notional and asset class. CEM was criticised for failing to recognise netting benefit in the add-on component and for being insufficiently risk-sensitive.

Layer 2 — Formula / Structure

CEM EAD = $\max(\text{MtM}, 0) + \text{Notional} \times \text{Add-on Factor (by asset class and maturity)}$. No netting benefit in add-on — significant overstatement for hedged books.

Layer 3 — Practitioner Usage

CEM was replaced by SA-CCR under Basel III, effective 2017 for most jurisdictions. The key weakness: CEM's add-on did not recognise that a long and short IRS in the same netting set partially offset each other's future exposure. SA-CCR corrected this by introducing the multiplier and aggregate add-on with partial netting. Understanding CEM is useful context for why SA-CCR was designed the way it was.

INTERVIEW
ANSWER

"CEM was the old standardised CCR method before SA-CCR — much simpler but much cruder. The critical weakness: the add-on component didn't recognise netting. If you had a EUR 100M pay-fixed IRS and a EUR 100M receive-fixed IRS with the same counterparty, CEM added both add-ons as if they were fully separate exposures. SA-CCR fixed this with the aggregate add-on formula that recognises partial netting benefit. Knowing CEM helps you explain why SA-CCR was designed as it was — the multiplier and partial netting are direct responses to CEM's failures."

CET1 — Common Equity Tier 1 [M1,M6]

Layer 1 — Plain English

The highest quality regulatory capital under Basel III — primarily retained earnings and paid-in ordinary share capital. The CET1 ratio is the primary measure of a bank's capital strength: CET1 capital divided by Risk-Weighted Assets.

Layer 2 — Formula / Structure

CET1 Ratio = CET1 Capital / RWA. G-SIB all-in CET1_req = 4.5% + 2.5% CCB + 1.0–3.5% G-SIB surcharge (+ SCB for US BHCs)

Layer 3 — Practitioner Usage

CET1 is the denominator of KVA: $E[K(t)] = E[RWA(t)] \times CET1_req$. A higher CET1 requirement means more capital rent per unit of RWA. Banks manage to a target CET1 above regulatory minimum to maintain a management buffer. For a Bucket 2 G-SIB the all-in CET1_req is 8.5% (without SCB) — raising the KVA cost of every trade by 21% vs a non-G-SIB.

INTERVIEW
ANSWER

"CET1 is the highest quality capital — mainly retained earnings and ordinary shares. The CET1 ratio is your primary capital health metric. For a G-SIB, the all-in requirement is the 4.5% minimum plus the conservation buffer plus the G-SIB surcharge, which can add up to 8.5% or more before the stress capital buffer. The KVA implication: the higher your CET1 requirement, the more capital you hold per unit of RWA, and the higher your KVA charge on every derivative trade. That's why G-SIBs are structurally disadvantaged vs non-G-SIBs on capital-intensive business."

Cherry-Picking — Cherry-Picking — Selective Enforcement of Derivative Contracts [M5]

Layer 1 — Plain English

The practice by which an administrator of a defaulted entity attempts to selectively enforce only those derivative contracts that are in-the-money for the estate, while disclaiming contracts requiring payment. ISDA close-out netting eliminates cherry-picking by requiring simultaneous termination of all contracts under the Master Agreement.

Layer 3 — Practitioner Usage

Cherry-picking was the central legal risk motivating the ISDA Master Agreement. The Hammersmith & Fulham case (1991) showed the catastrophic consequences when netting agreements failed. Post-ISDA, close-out netting has held in every major jurisdiction including the Lehman collapse (2008). Without close-out netting, banks must hold capital against gross rather than net exposure — potentially tripling RWA.

INTERVIEW ANSWER

"Cherry-picking is what an insolvency administrator would do without ISDA close-out netting: enforce the profitable contracts, disclaim the losing ones. ISDA eliminates this by requiring simultaneous termination of all trades under one Master Agreement, producing a single net close-out amount. The capital implication is enormous — regulators only allow net EAD for SA-CCR if there's a legally enforceable netting agreement. Without it, you're on gross EAD, which can be 3x higher. That's why netting opinions are non-negotiable."

Clean Risk — Clean Risk — Current Mark-to-Market Exposure [Risk]

Layer 1 — Plain English

The current mark-to-market credit exposure to a counterparty: the net present value of all open derivative positions, adjusted for any collateral posted or received. It represents the cost of replacing all trades if the counterparty defaulted today.

Layer 2 — Formula / Structure

$$\text{Clean Risk} = \text{MtM of open trades} - \text{Collateral held} + \text{Collateral posted (net, after eligible haircuts)}$$

Layer 3 — Practitioner Usage

Clean risk is the primary input to credit limit utilisation and forms the core of OBSI alongside PFE and settlement risk. For daily-margined CSA counterparties, clean risk is small (bounded by the MTA and threshold). For uncollateralised counterparties it can be very large. Real-time clean risk monitoring is the core function of credit control systems.

INTERVIEW ANSWER

"Clean risk is the current MtM cost of replacing all your trades with a counterparty if they defaulted right now — net of any collateral. It's the 'what would you lose today' number. Clean risk plus PFE gives you the pre-settlement risk profile, and together they form the basis of OBSI limit monitoring. In interviews: distinguish clean risk (today's number) from PFE (the potential future number) and from settlement risk (the in-the-pipe number). OBSI combines all three."

Close-out Netting — Close-out Netting — ISDA Section 6 [M5]

Layer 1 — Plain English

The legal mechanism under Section 6 of the ISDA Master Agreement that requires all transactions between two counterparties to terminate simultaneously upon a default event. A single net Close-out Amount replaces all individual gross obligations. This eliminates cherry-picking and enables net capital treatment.

🕒 Close-out Netting — Gross vs Net Exposure

WITHOUT NETTING (Gross):

Trade 1: +EUR 4.2M (counterparty owes dealer) → counts fully

Trade 2: -EUR 2.8M (dealer owes counterparty) → counts fully

Trade 3: +EUR 1.9M (counterparty owes dealer) → counts fully

Gross Exposure = $4.2 + 2.8 + 1.9 = \text{EUR } 8.9\text{M}$

WITH NETTING (Net):

Net Amount = $4.2 - 2.8 + 1.9 = \text{EUR } 3.3\text{M}$ (single close-out sum)

Capital reduction = $(8.9 - 3.3) / 8.9 = 63\%$

SA-CCR EAD: Net netting set → ~65% lower RWA than gross.

Layer 2 — Formula / Structure

Close-out Amount = Net of all terminated trade values at replacement cost (2002 ISDA Close-out Amount methodology)

Layer 3 — Practitioner Usage

Close-out netting reduces SA-CCR EAD by 50–70% vs gross treatment for diversified netting sets. Requires a current ISDA netting opinion confirming legal enforceability in the counterparty's jurisdiction. The 1992 ISDA dual methodology (Market Quotation or Loss) was replaced by the single 2002 Close-out Amount after Market Quotation failed in illiquid post-Lehman markets.

INTERVIEW ANSWER

"Close-out netting means that if a counterparty defaults, all trades under the ISDA Master terminate simultaneously and net to a single amount. You don't have to pay the losing trades and wait to receive on the winning ones — it's one net number. The capital benefit is huge: SA-CCR calculates EAD on the net position rather than the gross, which can reduce RWA by over 50%. The legal foundation is the ISDA netting opinion — without a current opinion confirming enforceability in the counterparty's jurisdiction, you lose the netting benefit and capital triples."

ColVA — Collateral Valuation Adjustment [M6]

Layer 1 — Plain English

The present value of the cheapest-to-deliver (CTD) optionality embedded in a multi-currency Credit Support Annex. The collateral poster has the right to deliver whichever eligible asset costs least — this choice has monetary value that must be priced into the trade. ColVA = 0 for single-currency CSAs.

Layer 2 — Formula / Structure

$$\text{ColVA} = -\int \text{EPE}(t) \times E[\max(s_{ij}(t), 0)] \times df(t) dt \quad \text{where } s_{ij} = \text{spread between eligible collateral funding rates}$$

Layer 3 — Practitioner Usage

ColVA is a benefit to the poster (long the CTD option) and a cost to the receiver (short). It grows with the number of eligible currencies, cross-currency basis volatility, and trade tenor. Hedged using cross-currency basis swaps. Under stress, ColVA faces a regime risk: the cheapest asset is often least deliverable in size — the option value is conditional on collateral actually being movable.

INTERVIEW ANSWER

"ColVA is the value of the right to post the cheapest collateral under a multi-currency CSA. Think of it like the CTD option in bond futures — the short always delivers the cheapest eligible bond. Here, the collateral poster always delivers the cheapest eligible asset. For a single-currency CSA, ColVA is zero — no choice, no option. For a EUR+USD+GBP CSA, you have a spread option on three rates. The K stands for Kapital — Collateral — sorry, the 'Col' is Collateral. The key practitioner insight: this option value is conditional on the collateral actually being movable in stress. When markets tighten, the cheapest asset is often the least deliverable in size."

Compression — Portfolio Compression — Notional Reduction

[M6,Risk]

Layer 1 — Plain English

The process of terminating multiple offsetting derivative transactions and replacing them with a smaller number of economically equivalent transactions, reducing gross notional without changing net market risk. Compression reduces RWA, leverage ratio exposure, and KVA.

Layer 2 — Formula / Structure

Compression ratio = (Gross Notional Before – Gross Notional After) / Gross Notional Before. KVA reduction $\approx$ proportional to EAD reduction from lower gross notional.

Layer 3 — Practitioner Usage

Multilateral compression (through TriOptima, Quantile, or LCH SwapClear) is more effective than bilateral compression because it can eliminate three-way or four-way cycles that bilateral compression cannot. Post-Basel III and post-UMR, compression has become a core balance sheet management tool — reducing gross notional reduces SA-CCR EAD (and therefore KVA), leverage ratio exposure, and aggregate SIMM IM (reducing MVA). Some estimates suggest EUR 100+ trillion of notional has been eliminated through compression since 2008.

INTERVIEW ANSWER

"Portfolio compression is the process of tearing up offsetting trades and replacing them with smaller equivalent net positions. The economics are unchanged — the market risk is the same — but you've dramatically reduced gross notional. Why does that matter? SA-CCR EAD is partly based on gross notional (through the effective notional in the add-on). Lower EAD means lower RWA means lower KVA. Lower gross notional also reduces leverage ratio exposure and SIMM IM (which reduces MVA). TriOptima and Quantile run multilateral compression cycles where multiple dealers eliminate circular flows simultaneously. It's a key balance sheet optimisation tool for any large derivatives dealer."

Concentration Risk — Concentration Risk [Risk]

Layer 1 — Plain English

The risk arising from excessive exposure to a single counterparty, sector, country, or correlated group. If that entity or group is stressed, the portfolio loss is disproportionate relative to the total book.

Layer 3 — Practitioner Usage

Managed through single-name limits (OBSI), sector limits, and country limits. The Archegos collapse (2021) exemplified concentration risk: multiple dealers each had large TRS exposures to the same handful of stocks. No single dealer saw the aggregate concentration — a fundamental information asymmetry of the bilateral framework. Post-Archegos, prime brokerage due diligence now includes mandatory leverage disclosure across counterparties.

INTERVIEW ANSWER

"Concentration risk is the danger of being too exposed to one name, sector, or geography. The Archegos example is perfect for interviews: each dealer had seemingly manageable bilateral exposure, but in aggregate across five dealers the concentration in four or five stocks was enormous. When prices fell, all dealers margin-called simultaneously, triggering a spiral. The lesson: bilateral limits don't protect against system-level concentration. That's the information asymmetry inherent in OTC markets."

CS01 — CS01 — Credit Spread 01 (1 Basis Point Credit Sensitivity)

[M3]

Layer 1 — Plain English

The change in value of a credit instrument for a 1 basis point parallel shift in the credit spread curve. CS01 measures sensitivity to credit spread movements — the credit analogue of DV01 for interest rates.

Layer 2 — Formula / Structure

$$\text{CS01} = \Delta \text{Value per 1bp shift in spread} = -dV/d\text{CDS_spread} \times 0.0001.$$

$$\text{For CVA: CS01_CVA} = d\text{CVA}/d\text{Spread} \times 0.0001$$

Layer 3 — Practitioner Usage

CS01 is the primary hedging sensitivity for CVA desks — used to size CDS hedge positions. FRTB SA-CVA requires reporting of CS01 sensitivities by counterparty, sector, and maturity bucket. KVA has a CS01 ($d\text{KVA}/d\text{Credit Spread}$) that is expensive under finite difference but tractable under AAD. A widening of 10 bps in a BBB counterparty's CDS spread increases CVA by approximately $\text{CS01} \times 10$.

INTERVIEW
ANSWER

"CS01 is your credit spread sensitivity — how much your position changes for a 1 basis point move in the credit spread. For a CVA desk, CS01 tells you how much CDS protection you need to hedge. If your CVA has a CS01 of EUR 50,000 per basis point and you want to hedge 100 bps of spread widening risk, you need EUR 5M of CDS protection. Under FRTB SA-CVA, you report these sensitivities by bucket to compute the regulatory capital charge."

CVA — Credit Valuation Adjustment [M3]

Layer 1 — Plain English

The adjustment to the risk-free price of a derivative that accounts for the possibility of counterparty default. CVA is the expected loss from counterparty default — the probability of default multiplied by the loss given default multiplied by the expected exposure at the time of default, integrated over the trade's life.

Layer 2 — Formula / Structure

$$\text{CVA} = -\int \text{LGD} \times \text{PD}(t) \times \text{EPE}(t) \times \text{df}(t) \, dt \quad (\text{unilateral simplified form})$$

Layer 3 — Practitioner Usage

CVA is mandatory under IFRS 13 for all derivative fair values. CVA desks hedge credit delta using CDS. Under FRTB, banks also hold capital against CVA risk (the risk that CVA marks change as spreads move). Wrong-way risk — where PD and EPE are positively correlated — amplifies CVA significantly. For fully collateralised (daily VM, zero threshold) trades, CVA approaches zero.

INTERVIEW
ANSWER

"CVA is the expected cost of counterparty default embedded in a derivative's price. The formula: integrate the probability of default times LGD times the expected positive exposure, discounted back to today. Three inputs, three handles: if the counterparty's CDS spread widens, PD goes up, CVA goes up, the CVA desk books a loss. If you add a VM CSA, EPE drops to near zero, CVA drops to near zero. The practical punchline: CVA is why adding a collateral agreement to an uncollateralised trade creates a large positive P&L event for the CVA desk — the reserve is released."

D

Default Fund — CCP Default Fund [M3]

Layer 1 — Plain English

A pool of pre-funded capital contributed by all clearing members of a CCP, used as the third line of defence in the loss waterfall following a member default. It absorbs losses that exceed the defaulting member's own IM and default fund contribution.

Layer 2 — Formula / Structure

Loss sequence: Defaulter VM → Defaulter IM → Defaulter DF → CCP Equity → Surviving Members DF

Layer 3 — Practitioner Usage

Default fund contributions are a regulatory capital cost for clearing members — they consume leverage ratio exposure and attract a small risk weight (2% for qualified CCPs under Basel). This cost is much lower than the bilateral CCR capital charge the clearing replaces. The allocation methodology (how losses are shared among surviving members) is a key design variable in CCP risk management.

INTERVIEW ANSWER

"The CCP default fund is the mutualised loss-absorbing pool contributed by all clearing members. It's the third line after the defaulter's own VM and IM. The key point interviewers want: default fund contributions have capital implications — they use leverage ratio capacity and carry a small risk weight even though it's much cheaper than bilateral capital. The systemic question is whether CCPs — by concentrating risk — create a new too-big-to-fail problem rather than solving the bilateral one."

Delta — Delta — First-Order Price Sensitivity [M0]

Layer 1 — Plain English

The change in value of a derivative for a one-unit move in the underlying price. Delta measures the linear price sensitivity and is the primary hedging metric for options and other non-linear instruments. Delta ranges from 0 to 1 for calls and -1 to 0 for puts.

Layer 2 — Formula / Structure

Delta = dV/dS . For European call: $\Delta = N(d1)$. For European put: $\Delta = N(d1) - 1$. For a linear instrument (IRS, forward): $\Delta = 1$.

Layer 3 — Practitioner Usage

Delta is the hedge ratio — a delta of 0.6 means you need to short 0.6 units of the underlying per unit of the derivative to be delta-neutral. Delta changes as the underlying moves (this rate of change is gamma). CVA has a credit delta — the sensitivity of CVA to counterparty CDS spread — which is the primary CVA hedging metric.

INTERVIEW ANSWER

"Delta is how much your derivative moves per one-unit move in the underlying. A call option with delta 0.6 gains EUR 0.60 for every EUR 1 rise in the underlying. To hedge, you short 0.60 of the underlying. As the underlying moves, delta changes — that rate of change is gamma, which tells you how often you need to rebalance your delta hedge. For an IRS, delta with respect to the underlying rate is essentially 1 — it's a linear instrument. The XVA application: CVA delta is $dCVA/dCDS$ spread, used to size the CDS hedge on the CVA book."

Dirty Risk — Dirty Risk — Total Credit Exposure Including Settlement [Risk]

Layer 1 — Plain English

The total credit exposure to a counterparty including both pre-settlement risk (clean risk) and settlement risk — the full exposure during the settlement pipeline. Dirty risk is highest on the settlement date of large transactions.

Layer 2 — Formula / Structure

Dirty Risk = Clean Risk + Settlement Risk (outstanding deliveries and payments awaiting confirmation)

Layer 3 — Practitioner Usage

Dirty risk spikes on settlement dates. A dealer that has sold EUR 100M of bonds but not yet received payment carries EUR 100M of dirty settlement risk. CLS (Continuous Linked Settlement) eliminated most FX settlement risk for its member currencies. For bonds and derivatives, settlement risk persists during the T+1 or T+2 window. Dirty risk is the correct limit utilisation measure during the settlement window.

INTERVIEW ANSWER

"Dirty risk is clean risk plus settlement risk — it's the total exposure including anything in the settlement pipeline. The classic scenario: you've executed a large bond sale, the counterparty has the bonds, you're waiting for cash. Until the cash settles, you have full settlement risk on that amount. That's why dirty risk is what you monitor on settlement dates, not clean risk. Herstatt risk — the 1974 German bank failure where USD was paid but DEM never received — is the original settlement risk event that drove creation of CLS."

Dodd-Frank — Dodd-Frank Wall Street Reform and Consumer Protection Act [M5]

Layer 1 — Plain English

The US federal law enacted in 2010 in response to the 2008 financial crisis, implementing the G20 OTC derivative reforms for US markets. Key provisions: mandatory CCP clearing for standardised swaps, SEF (Swap Execution Facility) trading, swap dealer registration, margin requirements, and swap data reporting.

Layer 3 — Practitioner Usage

Dodd-Frank is the US parallel to EMIR. The CFTC regulates swaps; the SEC regulates security-based swaps. US swap dealers must register with the CFTC and comply with margin rules implemented by the CFTC and prudential regulators. SIMM is accepted by US regulators for bilateral IM calculation. Cross-border equivalence determinations between CFTC and EMIR/MAS reduce duplicative compliance burdens for global dealers.

INTERVIEW ANSWER

"Dodd-Frank is the US post-crisis reform law that implemented the G20 OTC derivative commitments. Three key things for derivatives professionals: mandatory clearing for standardised swaps (through CFTC-registered DCOs), SEF trading requirements (electronic execution for liquid swaps), and the margin rules (CFTC and prudential regulator rules implementing UMR). The CFTC covers most swaps; the SEC covers security-based swaps like single-name CDS. Cross-border: CFTC has equivalence arrangements with EMIR and MAS so global dealers don't have to comply with two full regimes simultaneously."

DV01 — DV01 — Dollar Value of 01 (Interest Rate Sensitivity) [M0]

Layer 1 — Plain English

The change in value of a fixed income instrument or derivative for a 1 basis point parallel shift downward in interest rates. DV01 measures interest rate sensitivity and is the fundamental hedging metric for rate risk.

Layer 2 — Formula / Structure

$DV01 = -dP/dr \times 0.0001$. For an IRS: $DV01 \approx \text{Notional} \times \text{Supervisory Duration} \times 0.0001$. EUR 100M 5Y IRS $\approx$ EUR 4,500–5,000 DV01

Layer 3 — Practitioner Usage

DV01 is additive across a netting set — portfolio DV01 determines the size of the rate hedge required. KVA has an interest rate sensitivity ($dKVA/dRate$) computable via AAD. For a 5Y EUR 100M IRS, each basis point move in the yield curve changes the value by approximately EUR 4,500 — this drives daily VM calls as rates move.

INTERVIEW ANSWER

"DV01 is your interest rate sensitivity — how much the trade changes in value for a 1 basis point move in rates. For a EUR 100M 5-year IRS it's roughly EUR 4,500-5,000 per basis point. If rates move 10 bps against you, you're down roughly EUR 45,000-50,000 and that gets settled via a VM call. DV01 is additive, so you can sum it across a portfolio to find your net rate risk and size your hedge. It's also called PV01 in some contexts — same concept."

DVA — Debit Valuation Adjustment [M3]

Layer 1 — Plain English

The adjustment to derivative fair value reflecting the benefit arising from the reporting entity's own possibility of default. DVA is the mirror image of CVA from the counterparty's perspective — as your own credit deteriorates, DVA creates a mark-to-model profit.

Layer 2 — Formula / Structure

$$DVA = -\int LGD_{\text{own}} \times PD_{\text{own}}(t) \times ENE(t) \times df(t) dt \quad (\text{using own PD and Expected Negative Exposure})$$

Layer 3 — Practitioner Usage

DVA is required under IFRS 13 but deeply controversial — it implies a bank profits when its own credit deteriorates ("profit from impending doom"). Hedging DVA by buying protection on one's own default is legally and practically constrained. DVA P&L is closely scrutinised by auditors. CVA + DVA together are sometimes called bilateral CVA (bCVA).

INTERVIEW ANSWER

"DVA is your own credit adjustment — the benefit you receive because there's a chance you yourself might default and not have to pay. It's required under IFRS 13 but it's controversial because it means your fair value goes up when your credit deteriorates, which feels backwards. The practical challenge: you can't easily hedge it because buying protection on your own default is restricted. In interviews, flag the controversy and note that most banks include DVA in fair value but manage it separately from economic hedging."

E

EAD — EAD — Exposure at Default [M3]

Layer 1 — Plain English

The estimated exposure at the moment of counterparty default, used as the primary input to regulatory capital calculation. Under SA-CCR, $EAD = \alpha \times (RC + PFE)$ where $\alpha = 1.4$.

Layer 2 — Formula / Structure

$$EAD = 1.4 \times (RC + PFE) \quad | \quad RC = \max(V - C, 0) \quad | \quad PFE = \text{multiplier} \times \text{Aggregate Add-on} \quad | \quad RWA = EAD \times \text{Risk Weight}$$

Layer 3 — Practitioner Usage

EAD under SA-CCR is intentionally conservative — the alpha of 1.4 buffers against correlation between market and credit risk. VM reduces RC; IM reduces PFE via the multiplier. EAD is the bridge between market exposure and regulatory capital. For a netted portfolio, EAD is calculated on the net position — the capital benefit of netting flows directly through EAD.

**INTERVIEW
ANSWER**

"EAD under SA-CCR is 1.4 times the sum of replacement cost and potential future exposure. The replacement cost is the current MtM loss if the counterparty defaulted today — reduced by any VM held. The PFE is the potential future move — reduced by IM. Alpha of 1.4 adds a conservative buffer. EAD feeds directly into RWA: multiply by the counterparty's risk weight. For a BBB corporate that's 100%, so EAD = RWA in that case. The KVA formula then takes RWA × CET1_req to get the capital cost over the trade's life."

EMIR — EMIR — European Market Infrastructure Regulation [M5]

Layer 1 — Plain English

EU Regulation 648/2012 governing OTC derivatives in the European Union. Three pillars: mandatory CCP clearing for standardised OTC derivatives, bilateral margining for non-cleared trades, and reporting to a trade repository.

Layer 3 — Practitioner Usage

EMIR Refit (2024) updated reporting to CPMI-IOSCO critical data element standards. EMIR applies to EU financial and non-financial counterparties above clearing thresholds. Cross-border equivalence with CFTC (US) and MAS (Singapore) frameworks reduces duplicative compliance for global banks. EMIR is the EU parallel to Dodd-Frank — both implement the G20 Pittsburgh 2009 OTC derivatives commitments.

**INTERVIEW
ANSWER**

"EMIR is the EU's OTC derivatives regulation — three pillars: clearing obligation (standardised swaps must go to a CCP), bilateral margining (UMR-equivalent VM and IM requirements), and trade reporting (everything must be reported to a trade repository within one business day). EMIR Refit updated the reporting format. For a global bank, EMIR and Dodd-Frank overlap — both were implementing the G20 Pittsburgh commitments from 2009. The cross-border equivalence decisions between regulators determine whether a bank has to comply with both or can rely on one regime."

ENE — ENE — Expected Negative Exposure [M3]

Layer 1 — Plain English

The average of the negative mark-to-market values of a portfolio across future scenarios — the expected amount the reporting entity owes the counterparty at each future time point. ENE drives DVA and FBA (Funding Benefit Adjustment).

Layer 2 — Formula / Structure

$ENE(t) = E[\max(-MtM(t), 0)]$ averaged across scenarios. DVA uses
 $ENE: DVA = -\int LGD_{own} \times PD_{own} \times ENE(t) \times df(t) dt$

Layer 3 — Practitioner Usage

ENE is the mirror of EPE. Where EPE represents what the counterparty owes you, ENE represents what you owe them. FVA uses both — FCA (funding cost adjustment) uses EPE, FBA (funding benefit) uses ENE. In a symmetric bilateral relationship, ENE of party A equals EPE of party B.

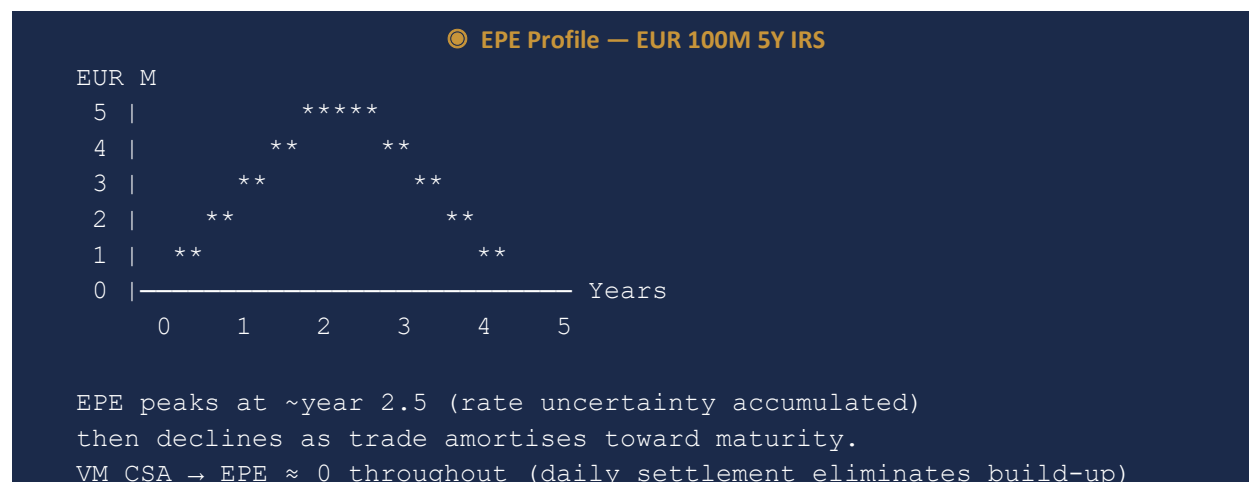
INTERVIEW ANSWER

"ENE is expected negative exposure — the average amount you owe the counterparty in future scenarios. EPE is what they owe you. DVA uses ENE — when your own credit deteriorates and ENE is large, DVA profit increases because you might not have to pay that negative exposure. FVA also uses ENE for the funding benefit side: when you're out of the money, you're effectively borrowing from the counterparty at risk-free, which is a funding benefit. The symmetry with EPE is important for understanding why bilateral CVA (CVA – DVA) nets out for perfectly symmetric trades."

EPE — EPE — Expected Positive Exposure [M3]

Layer 1 — Plain English

The average of the positive mark-to-market values of a portfolio across future market scenarios at each future time point. EPE is the primary exposure input to CVA, FVA, MVA, CoIVA, and KVA — it drives every valuation adjustment.



Layer 2 — Formula / Structure

$$\text{EPE}(t) = E[\max(\text{MtM}(t), 0)] \text{ averaged across market scenarios. } \text{CVA} = -\int \text{LGD} \times \text{PD}(t) \times \text{EPE}(t) \times df(t) dt$$

Layer 3 — Practitioner Usage

EPE for a 5Y EUR 100M IRS peaks around years 2–3 as rate moves accumulate, then declines toward maturity. EPE is sensitive to volatility (higher vol = higher EPE), wrong-way risk, and collateral regime. VM dramatically reduces EPE by settling accumulated MtM daily — a fully VM-collateralised trade has near-zero EPE and therefore near-zero CVA and FVA.

INTERVIEW ANSWER

"EPE is the expected positive exposure — the average of the positive MtM values across future market scenarios at each time horizon. It's the exposure profile that feeds into CVA, FVA, and KVA. The intuition: you simulate thousands of future market paths, average the positive values at each date, and integrate over the trade life. VM collapses EPE to near zero because daily settlement removes the accumulated positive exposure. That's why adding a VM CSA is the single biggest XVA event — it zeros out EPE and releases CVA and FVA reserves simultaneously."

ES — ES — Expected Shortfall (Conditional VaR) [M2]

Layer 1 — Plain English

A risk measure that quantifies the average loss in the worst X% of scenarios — the expected loss given that losses exceed the VaR threshold. Also called CVaR (Conditional Value at Risk) or ETL (Expected Tail Loss). ES replaced VaR as the primary FRTB market risk capital metric.

Layer 2 — Formula / Structure

$$\text{ES}(\alpha) = E[\text{Loss} \mid \text{Loss} > \text{VaR}(\alpha)] = (1/(1-\alpha)) \times \int \text{from } \alpha \text{ to } 1 \text{ of } \text{VaR}(u) du$$

Layer 3 — Practitioner Usage

ES at 97.5% over a liquidity-adjusted horizon is the FRTB IMA capital measure, replacing VaR at 99% over 10 days. ES is preferred because it is subadditive (portfolio ES ≤ sum of individual ES), captures tail severity

beyond the VaR threshold, and is a coherent risk measure. Backtesting ES is harder than backtesting VaR — FRTB relies on the P&L attribution test as the primary IMA validation tool.

INTERVIEW ANSWER

"Expected Shortfall answers a different question from VaR. VaR says: what is the loss threshold I won't exceed 99% of the time. ES says: given that I do exceed that threshold, what is my average loss? ES captures the severity of tail events, not just their frequency. FRTB switched from VaR to ES at 97.5% precisely because VaR ignores what happens in the worst 1% of cases. The coherence property — ES is subadditive, VaR is not — means portfolio ES is always $\leq$ the sum of individual ES, which makes it more suitable for aggregation."

F

FCA/FBA — FCA — Funding Cost Adjustment / FBA — Funding Benefit Adjustment ^[M4]

Layer 1 — Plain English

The two components of FVA, splitting the funding cost of positive exposure (FCA) from the funding benefit of negative exposure (FBA). FCA captures the cost of funding the uncollateralised EPE profile; FBA captures the benefit of being funded by the counterparty when the trade is out of the money.

Layer 2 — Formula / Structure

$$\text{FVA} = \text{FCA} + \text{FBA}. \quad \text{FCA} = -\int \text{EPE}(t) \times s_{\text{fund}} \times df(t) dt \text{ (cost)}. \quad \text{FBA} = +\int \text{ENE}(t) \times s_{\text{fund}} \times df(t) dt \text{ (benefit)}.$$

Layer 3 — Practitioner Usage

The FCA/FBA split addresses the double-count risk between FVA and DVA. DVA already captures the benefit of the reporting entity's own default on ENE positions — FBA captures the funding benefit of the same positions. Including both DVA and FBA double-counts the benefit. Most banks resolve this by either excluding DVA (keeping FBA) or excluding FBA (keeping DVA + FCA only). The choice has significant P&L implications for trades with large ENE profiles.

**INTERVIEW
ANSWER**

"FCA and FBA are the two halves of FVA. FCA is the funding cost adjustment — the cost of funding your positive exposure (EPE) at your funding spread above OIS. FBA is the funding benefit adjustment — the benefit you receive when the counterparty is effectively funding your negative exposure (ENE). The key interview point is the DVA/FBA double-count problem: DVA already captures the benefit of your potential default on ENE positions. If you also include FBA, you've captured that ENE benefit twice. Banks resolve this differently — some drop DVA and keep FCA+FBA, others keep DVA and only use FCA. The choice is significant and should be disclosed."

FRTB — FRTB — Fundamental Review of the Trading Book [M2]

Layer 1 — Plain English

The Basel Committee's comprehensive overhaul of market risk capital requirements for banks' trading books, finalised in 2019 and implemented from 2025. FRTB revised the trading/banking book boundary, replaced VaR with Expected Shortfall, overhauled the CVA capital framework, and tightened internal model approval.

Layer 3 — Practitioner Usage

FRTB's five key changes: (1) stricter trading/banking book boundary — instruments must be assigned at inception; (2) ES at 97.5% replacing VaR at 99%; (3) IMA approval at desk level — one underperforming desk can lose IMA without affecting others; (4) P&L attribution test as primary IMA validation; (5) new FRTB CVA framework (SA-CVA and BA-CVA replacing old CVA VaR). Most banks see higher trading book capital under FRTB.

**INTERVIEW
ANSWER**

"FRTB overhauled trading book capital — five key changes worth knowing: stricter book boundary so you can't cherry-pick where trades sit; ES replacing VaR; desk-level IMA approval; the P&L attribution test; and a new CVA capital framework. The last point is relevant for XVA roles — FRTB CVA introduced SA-CVA (which allows hedging recognition) and BA-CVA (basic, no hedging recognition). Banks that hedge CVA effectively benefit significantly from SA-CVA vs BA-CVA. Implementation is from 2025 in most jurisdictions."

FVA — FVA — Funding Valuation Adjustment [M4]

Layer 1 — Plain English

The adjustment to derivative fair value reflecting the cost or benefit of funding uncollateralised derivative exposures on the bank's balance sheet. FVA is zero for fully VM-collateralised trades and largest for long-dated uncollateralised positions.

Layer 2 — Formula / Structure

$$FVA = -\int \frac{EPE(t)}{dt} \times s_{fund} \times df(t) dt + \int \frac{ENE(t)}{dt} \times s_{fund} \times df(t) dt \text{ where } s_{fund} = \text{funding spread above OIS}$$

Layer 3 — Practitioner Usage

FVA is not mandatory under IFRS 13 but most major dealers include it on the grounds that a market participant would charge for it. FVA and CVA interact — double-count risk requires joint modelling (the FCA/FBA split). Under stress, FVA does not just widen — funding markets can close entirely, making the trade unfundable. FVA is zero for OIS-funded fully collateralised books.

INTERVIEW ANSWER

"FVA is the funding cost of carrying uncollateralised derivative exposure. If I have a EUR 100M uncollateralised IRS and my funding spread above OIS is 80 basis points, I'm paying 80 bps per year on the expected positive exposure to fund that position. That cost gets embedded in the trade price. The key insight: FVA is zero for a daily VM cash CSA — the trade is funded by the VM flows, not my balance sheet. Adding a CSA releases FVA, which is why collateral agreement renegotiations are major P&L events for XVA desks."

G**G-SIB — G-SIB — Global Systemically Important Bank** [M1,M6]**Layer 1 — Plain English**

A bank designated by the Financial Stability Board as globally systemically important, subject to additional CET1 capital surcharges of 1.0–3.5%, enhanced supervisory requirements, and resolution planning obligations.

Layer 2 — Formula / Structure

$$\text{G-SIB CET1}_{\text{req}} = 4.5\% \text{ min} + 2.5\% \text{ CCB} + \text{G-SIB surcharge (1.0-3.5\%)} = 8.0-10.5\% \text{ (before SCB)}. \text{ KVA G-SIB premium vs non-G-SIB} = \frac{\text{surcharge}}{\text{total CET1}_{\text{req}}}$$

Layer 3 — Practitioner Usage

G-SIB buckets are reassigned annually based on systemic importance scores across five categories. A Bucket 2 bank (1.5% surcharge) pays 21% more KVA than a non-G-SIB on the same trade. This differential is why G-SIBs structurally lose capital-intensive business to regional banks and why balance sheet optimisation (clearing, compression) is a strategic priority for G-SIB XVA desks.

INTERVIEW ANSWER

"G-SIBs are the globally systemically important banks — currently around 30 institutions — that carry an additional CET1 surcharge of 1–3.5% on top of the base requirements. The surcharge is meant to compensate for the 'too big to fail' problem: these banks have an implicit public backstop, which gives them a funding advantage, so they should pay a capital premium. For KVA: a Bucket 2 G-SIB has a CET1 requirement of 8.5% vs 7% for a non-G-SIB. That 21% higher capital requirement feeds directly into a 21% higher KVA on every trade. That's why G-SIBs push hard for clearing — it dramatically reduces the RWA on which KVA is charged."

Gamma — Gamma — Second-Order Price Sensitivity [MO]

Layer 1 — Plain English

The second derivative of a derivative's value with respect to the underlying price — the rate of change of delta as the underlying moves. Gamma measures the curvature of the value-price relationship. High gamma means the hedge ratio changes rapidly and requires frequent rebalancing.

Layer 2 — Formula / Structure

$$\text{Gamma} = d^2V/dS^2. \text{ For European options: } \text{Gamma} = N'(d1) / (S \times \sigma \times \sqrt{T}). \text{ Positive for long options, zero for linear instruments (futures, swaps)}$$

Layer 3 — Practitioner Usage

Gamma is critical for options books and for CVA desks where the credit delta changes as spreads move (CVA convexity). FRTB SA-CVA captures curvature via the curvature margin. Gamma trading — deliberately running a long gamma position to earn from volatility — is a common strategy for options dealers. Theta and gamma have an inverse relationship: paying theta is the cost of holding positive gamma.

INTERVIEW
ANSWER

"Gamma is the rate of change of delta — how quickly your hedge ratio changes as the underlying moves. A long option has positive gamma, which means as the underlying rises your delta increases, and as it falls your delta decreases — your position naturally gets longer in rising markets and shorter in falling markets. That's the convexity benefit you pay for via theta (time decay). For XVA, gamma matters because CVA has credit delta that changes as spreads widen — that second-order sensitivity is CVA convexity, which FRTB SA-CVA captures in the curvature margin."

Greeks — Greeks — Derivative Risk Sensitivities [M0]

Layer 1 — Plain English

The sensitivity measures that quantify how the value of a derivative changes with respect to changes in underlying market parameters. Named after Greek letters: Delta (underlying price), Gamma (delta sensitivity), Vega (volatility), Theta (time decay), Rho (interest rate).

Layer 2 — Formula / Structure

$$\text{Delta} = dV/dS \quad | \quad \text{Gamma} = d^2V/dS^2 \quad | \quad \text{Vega} = dV/d\sigma \quad | \quad \text{Theta} = dV/dt \quad | \quad \text{Rho} = dV/dr$$

Layer 3 — Practitioner Usage

Greeks are the primary language of derivative risk management. Every hedge ratio, risk limit, and XVA sensitivity is expressed in Greek terms. Under FRTB, regulatory capital sensitivities are calculated using prescribed delta, vega, and curvature (gamma proxy) for each risk factor. AAD computes all Greeks simultaneously in one backward sweep.

INTERVIEW
ANSWER

"The Greeks tell you how your derivative's value responds to market moves. Delta: how much it moves per 1 unit move in the underlying. Gamma: how quickly delta itself changes. Vega: sensitivity to implied volatility. Theta: daily time decay. Rho: sensitivity to interest rates. The practical use: if your delta is 0.6, you need to short 0.6 units of the underlying to be delta-hedged. As the underlying moves, gamma tells you how fast that ratio changes and how frequently you need to rebalance. FRTB uses these exact sensitivities to calculate market risk capital under the standardised approach."

H

Haircut — Haircut — Collateral Valuation Discount [M5]

Layer 1 — Plain English

A percentage reduction applied to the market value of collateral to buffer against the risk that its value falls before liquidation following a default. A 2% haircut on EUR 100M of government bonds means only EUR 98M of collateral value is recognised.

Layer 2 — Formula / Structure

$$\text{Adjusted Collateral Value} = \text{Market Value} \times (1 - \text{Haircut}\%). \text{ Net Collateral} = \sum \text{Adjusted Values across eligible assets}$$

Layer 3 — Practitioner Usage

Haircuts vary: 0% for cash, 0.5–8% for G10 government bonds (tenor-dependent), 6–12% for IG corporate bonds, 15–25% for equities. BCBS-IOSCO minimum haircut floors prevent compression to zero in benign markets. Procyclicality — compression in good times, spike in stress — amplified the 2008 crisis. Collateral transformation (IG bonds → government bonds) incurs an implicit haircut economics cost via the transformation fee.

INTERVIEW ANSWER

"A haircut is the discount applied to collateral value to protect the lender if the collateral's price falls before they can liquidate it. For government bonds it might be 2–3%, for equities 15–25%. The procyclicality problem is important for interviews: in good markets, competition drives haircuts toward zero. In stress, they spike to 20-25% simultaneously across all participants. That spike forces simultaneous collateral sales, depresses prices further, and triggers more sales — a classic collateral spiral. The BCBS-IOSCO minimum haircut floors try to put a floor under the good-time compression."

Headroom — Headroom — Available Limit Capacity [Risk]

Layer 1 — Plain English

The amount of additional exposure that can be taken with a counterparty before a credit limit is breached. $\text{Headroom} = \text{Limit} - \text{current utilisation}$. Real-time headroom monitoring determines whether new trades can be executed.

Layer 2 — Formula / Structure

Headroom = Limit - Current OBSI Utilisation. Pre-trade check:
Headroom post-trade = Limit - (Current OBSI + Estimated new trade OBSI) > 0

Layer 3 — Practitioner Usage

Pre-trade headroom checks are a regulatory expectation and internal control requirement. Systems must calculate headroom pre-trade (before booking) and post-trade (to confirm compliance). Large transactions may consume headroom across multiple sub-limits simultaneously. Proactive headroom management — trading to reduce exposure before limits are approached — is a core credit officer function.

INTERVIEW ANSWER

"Headroom is simply how much room you have left before hitting the limit — limit minus current utilisation. The practical importance: before every trade, the system should check pre-trade headroom to ensure the transaction won't cause a breach. Post-trade confirmation verifies the check was correct. For large trades that might consume significant headroom, a credit officer pre-approval is typically required. Managing headroom proactively — reducing exposure before limits are approached — is better practice than reacting to breaches after they occur."

Hurdle Rate — Hurdle Rate — Required Return on Equity [M6]

Layer 1 — Plain English

The minimum required return on equity capital set by a bank's board. The hurdle rate represents the cost of equity — the annual return shareholders require for providing capital. It is the h parameter in the KVA formula, converting capital requirements into an annual cost.

Layer 2 — Formula / Structure

KVA annual cost at time $t = h \times E[K(t)]$. Total KVA = $h \times \int E[K(t)] \times df(t) dt$. Typical range: 10–12% non-G-SIB, 12–15% G-SIB

Layer 3 — Practitioner Usage

A 1% change in hurdle rate changes KVA by approximately EUR 20,000 on a EUR 100M 5Y IRS (BBB, G-SIB Bucket 2). Post-crisis hurdles rose as equity investors repriced bank risk. Hurdle rates are bank-specific and confidential — they reflect the board's assessment of the minimum return needed to justify continued equity investment. A higher hurdle makes the bank more selective about capital-intensive trades.

INTERVIEW ANSWER

"The hurdle rate is the minimum return shareholders require on the capital they've provided — typically 10–15% for a bank. In KVA terms, it's the rent rate per unit of capital. If the hurdle is 12% and a trade requires EUR 260,000 of capital, the annual capital rent is EUR 31,200. Over five years, discounted, that's approximately EUR 50,000 of KVA. The insight for interviews: a lower hurdle means more trades pass the capital test — a higher hurdle means the bank is more selective. Post-crisis hurdles rose because equity investors demanded higher returns to compensate for new regulatory complexity."

IM — IM — Initial Margin [M4,M5]

Layer 1 — Plain English

Collateral posted to buffer against potential future exposure during the Margin Period of Risk — the time between the last VM exchange and full close-out and position replacement after a default. IM must be segregated under UMR and cannot be rehypothecated.

Layer 2 — Formula / Structure

IM sized at: 99th percentile of potential exposure change over MPoR. MPoR = 10 days bilateral, 5 days cleared. SIMM: $IM = \sqrt{[\Sigma (WS_i)^2] + \Sigma_{i \neq j} (\rho_{ij} \times WS_i \times WS_j)}$

Layer 3 — Practitioner Usage

IM reduces SA-CCR PFE (via the multiplier), lowering KVA. But IM must be funded (MVA increases). The net effect depends on relative sizes. IM is protected in insolvency — segregated accounts return to the poster even if the receiver defaults. The transformation fee for non-eligible assets adds to all-in MVA cost.

INTERVIEW ANSWER

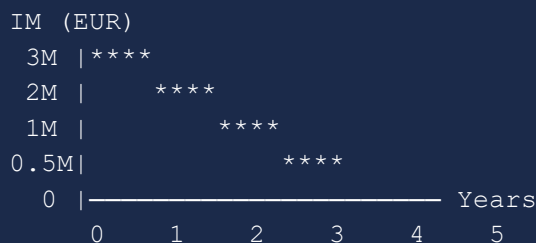
"Initial Margin buffers against potential future exposure during the close-out period — the 10 days after a default when you're replacing all the positions. VM settles what has already happened; IM covers what might happen in those 10 days. Under UMR it must be segregated — held by a third-party custodian, can't be reused by the receiver. That segregation is the source of MVA: the locked collateral costs the poster their funding spread. The interaction with KVA: IM reduces SA-CCR EAD (which reduces KVA) but you pay MVA to fund it. The net depends on which is larger."

IM Amortisation — IM Amortisation — Margin Decay Over Trade Life [M4,M5]

Layer 1 — Plain English

The phenomenon where the Initial Margin requirement for a trade declines as it amortises toward maturity, because the forward-looking risk profile (and therefore SIMM IM) shrinks as remaining tenor decreases. Quants modelling MVA must account for this decay rather than assuming flat IM over the trade life.

IM Amortisation — Schematic Decay Profile



Flat IM assumption (wrong):

Amortising IM (correct): *****

Flat assumption overstates MVA significantly.

Correct MVA uses declining $E[\text{SIMM_IM}(t)]$ path.

Layer 2 — Formula / Structure

$$\text{MVA} = -\int_0^T E[\text{SIMM_IM}(t)] \times s_{\text{fund}} \times df(t) dt \quad \text{where } E[\text{SIMM_IM}(t)] \text{ declines with } t \text{ as residual tenor decreases.}$$

Layer 3 — Practitioner Usage

The SIMM IM for a 5Y IRS at inception reflects 5 years of future risk. One year later, the same trade has only 4 years remaining — SIMM IM is lower. Failing to model this decay (using inception IM as a flat profile) systematically overstates MVA. The decay shape depends on the trade type: for an IRS, IM declines roughly linearly with residual tenor. For options, IM decays faster as vega and gamma decline near expiry. Accurate MVA modelling requires simulating the full IM profile path, not just the inception value.

INTERVIEW ANSWER

"IM amortisation — or margin decay — is the observation that Initial Margin requirements fall as a trade approaches maturity. At inception of a 5Y IRS, SIMM IM reflects 5 years of potential rate moves. At year 4, there's only 1 year left — SIMM IM is roughly 80% lower. If you model MVA using the inception IM as a flat profile over 5 years, you're overcharging significantly. The correct approach simulates the full $E[\text{SIMM_IM}(t)]$ path — typically declining roughly with residual tenor for vanilla trades. For exotic products with non-linear payoffs, the decay can be non-monotonic. This is an active area of MVA modelling sophistication at XVA desks."

IMM — IMM — Internal Model Method [M1,M3]

Layer 1 — Plain English

A Basel-approved approach allowing banks to use their own internal simulation models to calculate EAD for CCR, rather than the standardised SA-CCR approach. IMM typically produces lower RWA than SA-CCR for diversified, well-hedged derivative books. Subject to the Basel IV output floor (72.5% of SA RWA).

Layer 2 — Formula / Structure

IMM EAD = $\alpha \times E[\text{EPE}]$ from internal simulation, subject to regulatory floor: Effective RWA = $\max(\text{IMM RWA}, 72.5\% \times \text{SA-CCR RWA})$

Layer 3 — Practitioner Usage

IMM approval requires regulatory validation of the simulation model, stress testing methodology, and governance framework. Once approved, banks compute EAD using Monte Carlo simulation of market scenarios, producing potentially much lower RWA than SA-CCR for hedged books. The Basel IV output floor significantly constrains this benefit — banks cannot go below 72.5% of SA-CCR-equivalent RWA regardless of model sophistication. Many banks have found that the cost of maintaining IMM approval is no longer justified given the output floor.

INTERVIEW
ANSWER

"IMM is the internal model approach for CCR — banks use their own simulation models rather than SA-CCR's standardised formula. The benefit: a well-hedged book can show much lower EAD under IMM than SA-CCR because the model captures actual netting benefit and correlation effects. The catch: Basel IV output floor of 72.5% means you can't go below 72.5% of SA-CCR anyway. At that point, the cost of maintaining a validated IMM (model governance, regulatory approval, ongoing backtesting) may exceed the capital benefit. Many banks are questioning whether to maintain IMM approval post-Basel IV. The answer depends on whether 72.5% of SA-CCR is still better than SA-CCR — which depends on portfolio composition."

IRB — IRB — Internal Ratings-Based Approach [M1]

Layer 1 — Plain English

A Basel framework allowing banks to use their own internal credit rating models to calculate credit risk RWA for the banking book, rather than the standardised approach (SA) risk weights. IRB produces risk weights based on internal estimates of PD, LGD, and EAD for each exposure.

Layer 2 — Formula / Structure

IRB RWA = $f(\text{PD}, \text{LGD}, \text{M}, \text{correlation}) \times \text{EAD}$. For corporate exposures: $\text{risk weight} = \text{LGD} \times N[G(\text{PD})/\sqrt{(1-R)} + \sqrt{(R/(1-R))} \times G(0.999)] \times \text{maturity adjustment}$

Layer 3 — Practitioner Usage

IRB produces lower RWA than SA for high-quality (low PD) borrowers — a key source of capital efficiency for banks with strong credit risk models. Basel IV output floor (72.5% of SA RWA) constrains IRB benefit, particularly for low-risk portfolios where IRB produced the largest SA discount. IRB is not directly applicable to derivative CCR (which uses SA-CCR or IMM) but affects the overall capital stack that determines KVA through the bank's aggregate CET1 ratio.

INTERVIEW
ANSWER

"IRB allows banks to use their own models to set credit risk capital rather than Basel's standardised risk weights. A high-quality corporate borrower might get a 20-30% risk weight under IRB vs 100% under SA — a massive capital saving. Basel IV output floor constrains this: if IRB produces less than 72.5% of SA RWA, the floor kicks in. For derivatives roles, IRB matters less directly (CCR uses SA-CCR or IMM) but understanding that the bank's overall CET1 ratio and capital strategy is shaped by IRB helps contextualise where KVA fits in the broader capital framework."

ISDA Master — ISDA Master Agreement [M5]

Layer 1 — Plain English

The standardised bilateral contract published by the International Swaps and Derivatives Association that governs all OTC derivative transactions between two counterparties. It provides for close-out netting on default, standard events of default, termination events, and governing law. Two versions remain in use: ISDA 1992 and ISDA 2002.



Layer 3 — Practitioner Usage

ISDA 1992 used dual close-out methodology (Market Quotation or Loss) that failed in illiquid Lehman markets. ISDA 2002 introduced a single Close-out Amount — more robust under stress. The Master cannot exist without a Schedule (which configures it). The CSA is optional — and cannot exist without a Master. Both versions remain in active use; 2002 is preferred for new relationships.

INTERVIEW ANSWER

"The ISDA Master Agreement is the standardised legal framework that governs all OTC derivative trades between two parties. It does three critical things: provides for close-out netting on default (eliminating cherry-picking), standardises events of default, and enables regulatory capital netting benefit. The 1992 version had two close-out methods — Market Quotation and Loss — and Market Quotation failed in illiquid post-Lehman markets. The 2002 version fixed this with a single Close-out Amount. Key dependency: the Schedule is part of the Master (inseparable), the CSA is optional but cannot exist without a Master."

Issuer Risk — Issuer Risk — Credit Risk from Bond/Note Holdings

[Risk]

Layer 1 — Plain English

Credit risk arising from holding a security issued by a counterparty — bonds, notes, or structured products — where the issuer's default would cause loss of principal and/or coupon. Distinct from CCR (derivative contract risk).

Layer 3 — Practitioner Usage

Issuer risk and CCR are typically managed against a combined single-name credit limit to prevent aggregate concentration if that counterparty defaults. Under Basel, issuer risk in the banking book uses the credit risk framework (IRB or SA) rather than SA-CCR. The combined limit is the correct control for preventing a bank from simultaneously holding large derivative and bond exposures to the same name.

INTERVIEW ANSWER

"Issuer risk is the credit risk from holding bonds or other securities issued by a counterparty — separate from the derivative exposure to the same name. The reason they need to be managed together: if a counterparty defaults, both the derivative MtM exposure and the bond principal are at risk simultaneously. That's why banks set combined limits spanning both issuer and CCR exposure. The key distinction: derivative exposure is bilateral and uncertain (could be in or out of the money); bond exposure is one-directional (you hold the bond, issuer owes you)."

K

KVA — KVA — Capital Valuation Adjustment (Kapital) [M6]

Layer 1 — Plain English

The present value of the cost of holding regulatory capital against a derivative trade over its entire life. The "K" stands for "Kapital" — the German spelling — adopted from the Basel framework to avoid collision with CVA's "C" (Credit) for Capital. KVA is the rent paid on regulatory capital for the privilege of holding a derivative on the balance sheet.

Layer 2 — Formula / Structure

$$KVA = h \times \int_0^T E[K(t)] \times df(t) dt \quad | \quad E[K(t)] = E[RWA(t)] \times CET1_{req}$$

$$| \quad E[RWA(t)] = 1.4 \times (RC + E[PFE(t)]) \times RW$$

Layer 3 — Practitioner Usage

Pre-Basel III KVA was negligible — capital was cheap and abundant. Post-Basel III, with G-SIB CET1 requirements of 8.5–13%+, the annual capital cost on a EUR 100M IRS exceeds EUR 10,000–50,000 depending on counterparty quality and G-SIB bucket. Under stress, KVA does not reprice — it jumps, as buffers bind and SA-CCR inflates. At that point KVA becomes a shadow price of admissibility, not a cost adjustment.

INTERVIEW ANSWER

"KVA is the cost of regulatory capital — the rent you pay for occupying balance sheet space. The K is from Kapital, the German spelling, borrowed because C was already taken by CVA. The formula: hurdle rate times the integral of expected capital over the trade's life. The practical punchline: for a EUR 100M 5-year IRS with a BBB corporate at a G-SIB Bucket 2 bank, KVA is roughly EUR 49,000. A non-G-SIB would pay EUR 41,000 for the same trade — 18% less. That differential means G-SIBs structurally lose capital-heavy business to regional banks. The stress insight to flag: KVA doesn't drift up in stress — it jumps when buffers bind. That's a regime change, not a repricing."

L

Leverage Ratio — Leverage Ratio — Non-Risk-Sensitive Capital Backstop [M1,M6]

Layer 1 — Plain English

A non-risk-sensitive capital measure that constrains total balance sheet assets relative to Tier 1 capital, regardless of risk weights. The leverage ratio prevents banks from using low-risk-weight assets to build excessive gross leverage even when RWA-based ratios look comfortable.

Layer 2 — Formula / Structure

Leverage Ratio = Tier 1 Capital / Total Exposure Measure (gross, limited netting). Minimum: 3% globally; 3.5–5% for G-SIBs

Layer 3 — Practitioner Usage

For derivative-heavy banks, the leverage ratio exposure measure includes gross derivative notional (with adjustments) rather than net SA-CCR EAD. This means cleared IRS trades — which have low RWA but large gross notional — can be constrained by the leverage ratio before the RWA-based capital requirement. When LR binds, KVA must use LR capital cost — typically higher than RWA-based KVA.

INTERVIEW ANSWER

"The leverage ratio is the non-risk-sensitive backstop — Tier 1 capital divided by gross exposure, ignoring risk weights. It prevents a bank from holding enormous notional with tiny capital just because the risk weights are low. For derivatives, the exposure measure uses gross notional (with adjustments) rather than the net SA-CCR EAD — so even a fully hedged cleared book consumes leverage capacity. When the leverage ratio binds rather than the RWA ratio, KVA is calculated on leverage ratio capital, which is typically higher. That's why some trades that look cheap on an RWA basis are actually expensive on a total balance sheet basis."

LGD — LGD — Loss Given Default [M3]

Layer 1 — Plain English

The proportion of exposure that is actually lost when a counterparty defaults, expressed as a percentage of the exposure at default. $LGD = 1$ minus the recovery rate. For unsecured OTC derivatives, Basel regulatory LGD is typically 60% (implying 40% recovery).

Layer 2 — Formula / Structure

$$CVA = -\int LGD \times PD(t) \times EPE(t) \times df(t) dt. \quad LGD = 1 - \text{Recovery Rate.}$$

Regulatory LGD for derivatives: typically 45–60% depending on seniority and collateralisation

Layer 3 — Practitioner Usage

LGD is a key CVA input. Lower LGD (better recovery) reduces CVA. Collateralised trades have lower effective LGD — the VM/IM held at default reduces the net loss. Wrong-way LGD occurs when collateral value falls precisely when the counterparty defaults — amplifying losses. Market-implied LGD can be

derived from CDS: $PD \times LGD = \text{CDS spread (simplified)}$, but LGD is typically assumed (40–60%) rather than independently estimated.

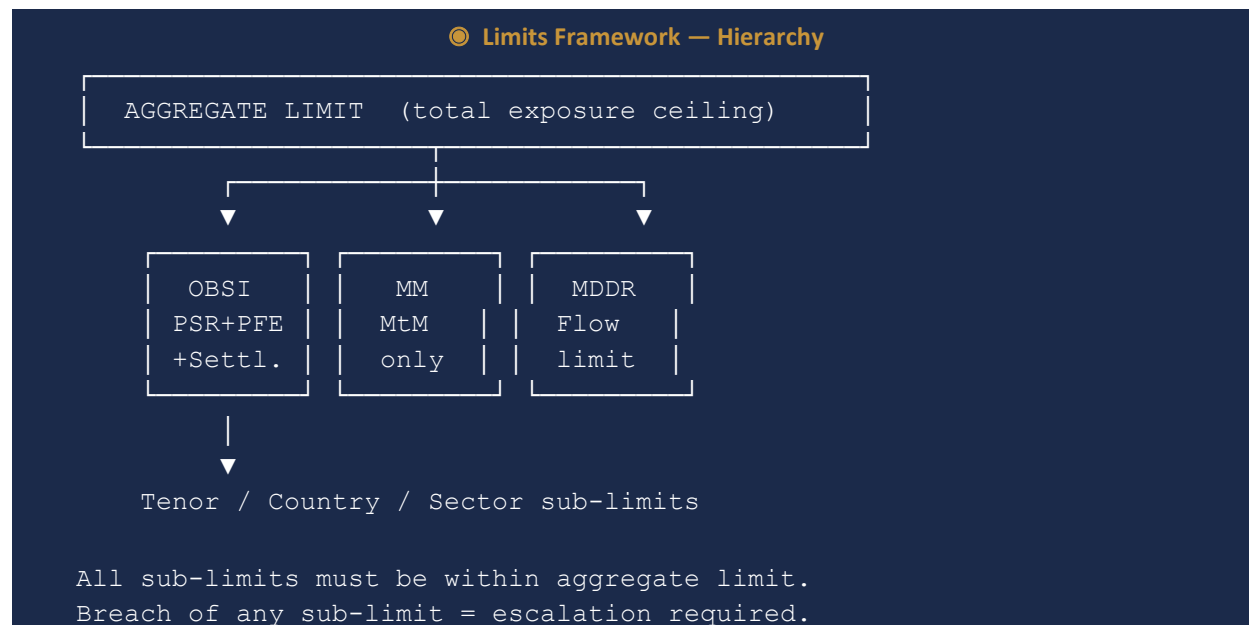
INTERVIEW ANSWER

"LGD is the fraction of exposure you actually lose in a default — 1 minus the recovery rate. For unsecured OTC derivatives, Basel assumes 60% LGD (40% recovery). Collateral improves LGD — if you hold VM equal to the MtM, your effective LGD approaches zero for that portion. The CVA formula multiplies PD by LGD by EPE — so if LGD halves, CVA halves. In interviews: distinguish market-implied LGD (derived from CDS spreads with an assumed PD) from accounting-based LGD (internal credit models). CVA desks typically use market-implied LGD via the CDS market."

Limits Framework — Limits Framework — Credit and Market Risk Limit Hierarchy [Risk]

Layer 1 — Plain English

The structured, multi-dimensional hierarchy of credit and market risk limits that constrains exposure to counterparties, markets, and products. A complete limits framework combines aggregate limits, product limits (OBSI, MM, MDDR), tenor limits, country limits, and sector limits.



Layer 3 — Practitioner Usage

Limits are approved by the credit committee for defined tenors and reviewed at least annually — or immediately following material credit events. Real-time limit monitoring is a regulatory expectation. The

framework is designed to be multi-dimensional: a counterparty can breach its OBSI limit (total exposure) while remaining within its MM limit (current MtM) and vice versa.

**INTERVIEW
ANSWER**

"A limits framework is a multi-dimensional control system. At the top: aggregate limit — total exposure across everything. Below that: product limits like OBSI (the main pre-settlement risk limit), MM (current MtM only), and MDDR (daily deal rate). Then tenor limits (no trades beyond X years), country limits (total exposure to counterparties domiciled in a given country), and sector limits (e.g. total financial institution exposure). The key insight for interviews: limits are prospective controls — they constrain what you can do, not just measure what you've done. Real-time monitoring is what makes them operational."

M

MDDR — MDDR — Maximum Daily Deal Rate [Risk]

Layer 1 — Plain English

A limit on the volume or notional of new transactions that can be executed with a counterparty in a single trading day. MDDR controls the pace at which new exposure builds up, preventing concentration of new business in a short window.

**Layer 3 — Practitioner
Usage**

MDDR is a flow limit (constrains the rate of new business) rather than a stock limit (constrains total outstanding). Particularly relevant as counterparties approach their aggregate limit headroom — MDDR determines how quickly remaining headroom is consumed. MDDR breaches require credit officer approval before new transactions can proceed.

**INTERVIEW
ANSWER**

"MDDR is a flow limit — it controls how fast you can build exposure with a counterparty in a single day. Imagine a counterparty with EUR 500M remaining aggregate limit headroom but USD 300M MDDR. You can book up to USD 300M equivalent in a day, not more. This prevents a trader from using up all available headroom in one large trade that the credit officer might not have reviewed. It's a governance control as much as a risk control — it ensures credit oversight keeps pace with trading activity."

MM Limit — MM Limit — Mark-to-Market Exposure Limit [Risk]

Layer 1 — Plain English

A credit limit that constrains the current mark-to-market exposure to a counterparty — the net present value of what they currently owe, before accounting for potential future moves. MM is the current cost of replacing all outstanding transactions if the counterparty defaulted today.

Layer 2 — Formula / Structure

$$\text{MM Utilisation} = \frac{\text{Net positive MtM (in-the-money trades, net of collateral held)}}{\text{MM Limit}}$$

Layer 3 — Practitioner Usage

MM limits sit alongside PFE limits within the OBSI framework. For daily-margined CSA counterparties MM exposure is typically small (bounded by MTA and threshold). For uncollateralised counterparties it can be very large. An MM breach means current exposure exceeds the approved threshold — may require immediate collateral call or trade reduction.

INTERVIEW ANSWER

"The MM limit controls how large your current MtM exposure can be to a counterparty — the replacement cost right now if they defaulted. It's a snapshot: no potential future moves included. For collateralised counterparties the MM stays near zero because VM settles it daily. For uncollateralised ones it can build significantly with market moves. The distinction from OBSI: MM is the current number, OBSI includes the future potential (PFE) and settlement pipeline too. You can have a small MM but large OBSI on a long-dated trade with a large PFE."

MPoR — MPoR — Margin Period of Risk [M4,M5]

Layer 1 — Plain English

The time from the last variation margin exchange to the point where all positions with a defaulted counterparty have been fully closed out and replaced. The MPoR determines how much potential future exposure can accumulate before the surviving party has fully hedged — and therefore drives the size of Initial Margin.

Layer 2 — Formula / Structure

**IM sized at: 99th percentile potential exposure change over MPoR.
Regulatory floors: 10 business days (bilateral non-cleared), 5
business days (centrally cleared)**

Layer 3 — Practitioner Usage

MPoR is extended by 10 business days if VM is called less frequently than daily. In stress scenarios (Lehman-type systemic event), actual MPoR can extend to 20–30 days. SA-CCR uses MPoR as an input to the supervisory maturity factor. A longer MPoR requires more IM and increases MVA.

INTERVIEW ANSWER

"MPoR is the time from your last VM exchange to when you've fully replaced all positions with a defaulted counterparty. During that window markets keep moving but you're not collecting VM. That's the gap IM is designed to cover. The regulatory floor is 10 days bilateral, 5 days cleared — the reasoning being that CCP standardised procedures are faster than bilateral close-out. In stress, 10 days is often too optimistic — Lehman took much longer. That's why stress testing uses 20-30 day MPoR assumptions. For MVA, longer MPoR means larger IM means higher funding cost."

MTA — MTA — Minimum Transfer Amount [M4,M5]

Layer 1 — Plain English

The minimum size of a collateral call or return under a CSA. Collateral movements smaller than the MTA are not made. The MTA prevents operationally burdensome small transfers but creates a band of uncollateralised exposure that persists between VM calls.

Layer 2 — Formula / Structure

**VM call is made only if: $|\text{Delivery Amount}| > \text{MTA}$. Residual
uncollateralised exposure $\approx \text{MTA} / 2$ on average between calls
(random walk approximation).**

Layer 3 — Practitioner Usage

MTA interacts with the Threshold to define the total permanently uncollateralised band. For CVA and FVA modelling, MTA creates a residual EPE even for CSA counterparties — it must be explicitly modelled rather than assuming perfect collateralisation. Typical MTAs: EUR 100,000–500,000 for bank-to-bank; EUR 1–5M for corporate counterparties. Regulatory guidance (BCBS-IOSCO) sets maximum MTA of EUR 500,000 for UMR in-scope entities.

INTERVIEW
ANSWER

"MTA is the minimum transfer amount — the floor size for any collateral call. If the MTA is EUR 500,000 and the VM call is only EUR 300,000, the call is not made. This prevents small, operationally expensive transfers but creates a permanently uncollateralised band up to the MTA size. For CVA and FVA modelling this matters: you can't assume perfect collateralisation just because a CSA exists. The MTA creates residual EPE that must be modelled explicitly. BCBS-IOSCO caps MTA at EUR 500,000 for UMR in-scope relationships. Always check both the Threshold and MTA when assessing how much residual XVA remains under a CSA."

MtM — MtM — Mark-to-Market [M3,M5]

Layer 1 — Plain English

The practice of valuing a financial instrument at its current market price rather than its original cost or face value. The MtM value represents what it would cost to replace the instrument in the current market — and is the basis for daily VM calls.

Layer 2 — Formula / Structure

MtM (IRS, pay-fixed) = PV(floating leg) - PV(fixed leg), discounted at OIS curve. Positive MtM = in-the-money for the dealer

Layer 3 — Practitioner Usage

Daily MtM is the basis for VM calls. As rates move, the IRS MtM changes and VM is exchanged to settle the accumulated gain/loss. MtM disputes arise on illiquid products where models diverge. MtM is the RC component of SA-CCR EAD. Under IFRS 13, derivatives must be held at MtM fair value including XVA adjustments.

INTERVIEW
ANSWER

"Mark-to-market means valuing the trade at today's market price — what would it cost to replace it right now. For an IRS, that's the present value of the floating leg minus the present value of the fixed leg, discounted at OIS. If rates have moved in your favour since inception, MtM is positive and the counterparty owes you VM. If rates moved against you, MtM is negative and you owe them VM. The daily VM settlement cycle means that accumulated MtM is regularly reset to near zero — which is why adding a VM CSA dramatically reduces CVA and FVA."

MVA — MVA — Margin Valuation Adjustment [M4]

Layer 1 — Plain English

The present value of the funding cost of posting Initial Margin over a derivative trade's life under UMR. MVA arises because IM posted to segregated accounts cannot be rehypothecated — the locked collateral costs the poster their full funding spread.

Layer 2 — Formula / Structure

$$MVA = -\int SIMM_IM(t) \times s_fund \times df(t) dt \quad \text{where } SIMM_IM(t) = \text{expected IM profile, } s_fund = \text{funding spread above OIS}$$

Layer 3 — Practitioner Usage

MVA is driven by the SIMM IM amount, the funding spread, and trade tenor. MVA interacts with KVA: IM reduces SA-CCR PFE (lowering KVA) but must be funded (increasing MVA). SIMM recalibrations directly change MVA. New trades that net against existing positions reduce aggregate SIMM IM and therefore reduce MVA.

INTERVIEW ANSWER

"MVA is the funding cost of posting Initial Margin. Under UMR, IM goes into a segregated account — you can't use those bonds or cash for anything else. That lockup has a cost equal to your funding spread times the IM amount times the life of the trade. For a EUR 100M IRS, SIMM IM might be EUR 2–3M locked up for 5 years. At an 80 bps funding spread, that's EUR 16,000–24,000 per year in MVA. The interaction with KVA: IM reduces your SA-CCR PFE, which lowers KVA. But MVA goes up. Whether the net is positive or negative depends on the specific trade and collateral regime."

N

Netting Opinion — Netting Opinion — Legal Enforceability Confirmation [M5]

Layer 1 — Plain English

A formal legal opinion from local counsel confirming that the close-out netting provisions of the ISDA Master Agreement are enforceable under a specific jurisdiction's insolvency law. Without a current netting opinion, regulators do not permit netting benefit for capital calculation.

Layer 3 — Practitioner Usage

ISDA publishes netting opinions for 70+ jurisdictions. Banks must maintain current opinions for every counterparty jurisdiction. An expired or absent opinion means gross EAD must be used for SA-CCR — potentially tripling the capital charge vs netted EAD. Netting opinions require periodic renewal as local insolvency laws change. This is a non-trivial legal infrastructure cost, particularly for banks with counterparties in emerging market jurisdictions.

INTERVIEW ANSWER

"A netting opinion is the legal document that confirms close-out netting is actually enforceable in the counterparty's jurisdiction. Without it, you can't use net EAD — you're on gross, which can triple your capital requirement. ISDA publishes these for 70+ jurisdictions but banks must make sure they're current. The practical interview point: the benefit of netting is entirely dependent on legal enforceability. The most sophisticated SA-CCR model in the world delivers zero benefit if the netting opinion has lapsed or doesn't cover the jurisdiction. This is why legal infrastructure — not just quantitative infrastructure — is critical in a derivatives business."

Netting Set — Netting Set — SA-CCR Unit of Calculation [M5]

Layer 1 — Plain English

All derivative transactions between two counterparties covered by a legally enforceable bilateral netting agreement. The netting set is the fundamental unit for SA-CCR EAD calculation — all trades within a set produce a single net EAD.

Layer 2 — Formula / Structure

$$\text{EAD (netting set)} = 1.4 \times (\text{RC}_{\text{net}} + \text{PFE}_{\text{aggregate}}).$$

Compare: Gross EAD = $\sum$ individual trade EADs. Netting benefit = $(\text{Gross EAD} - \text{Net EAD}) / \text{Gross EAD}$

Layer 3 — Practitioner Usage

The capital benefit of netting can reduce EAD by 50–70% vs gross for diversified portfolios. Netting sets are bounded by the ISDA Master — trades under different Masters are different netting sets. Cleared trades form separate netting sets at each CCP. A counterparty with two separate ISDA Masters (common for legacy book management) has two separate netting sets.

**INTERVIEW
ANSWER**

"A netting set is everything covered by one ISDA Master Agreement — all those trades net to a single EAD under SA-CCR. The benefit can be enormous: a portfolio with EUR 6.4M gross positive exposure and EUR 3.4M gross negative exposure has a net of EUR 3M — a 53% reduction in credit exposure. That feeds into a 60–70% reduction in SA-CCR EAD and therefore RWA. The key constraint: different ISDA Masters = different netting sets, no cross-netting. And cleared trades sit in a CCP netting set, not your bilateral one."

O**OBSI — OBSI — Outstanding Business and Settlement Indicator****[Risk]****Layer 1 — Plain English**

The primary composite credit risk limit metric that captures the full scope of pre-settlement exposure: current MtM (replacement cost), potential future exposure (PFE add-on), and any outstanding settlement obligations. $OBSI = MtM + PFE + Settlement\ Risk$.

◎ OBSI — Components of Pre-Settlement Risk

$$OBSI = MTM + PFE + Settlement\ Risk$$

MTM	Current replacement cost (net of collateral) = "what you'd lose if default happened today"
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PFE	Potential Future Exposure (95th pctlile) = "what exposure might grow to in future"
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SETTL	Settlement Risk (in-transit payments/bonds) = "what's in the pipeline, not yet confirmed"
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$$Utilisation\ \% = OBSI / OBSI\ Limit \times 100$$

**Layer 2 — Formula /
Structure**

$$\text{OBSI} = \text{MTM exposure (net, after collateral)} + \text{PFE (potential future exposure)} + \text{Settlement Risk (pending deliveries/payments not yet confirmed)}$$

Layer 3 — Practitioner Usage

OBSI is the most comprehensive single-number representation of credit exposure to a counterparty at any point. It combines the current replacement cost, the potential future build-up, and the settlement pipeline. Real-time OBSI monitoring against approved limits is the core function of credit control systems. Limit utilisation = $\text{OBSI} / \text{OBSI Limit}$.

INTERVIEW ANSWER

"OBSI is the main pre-settlement risk limit metric — it captures everything: current MtM, potential future exposure via PFE, and settlement risk from trades in the pipeline. It's the three-in-one number. MtM is what you'd lose today. PFE is what you might lose in future as markets move. Settlement risk is what's in transit. The sum is OBSI — the total pre-settlement credit exposure. Monitoring OBSI in real-time against the approved limit is the primary job of a credit controller in a derivatives business."

OIS — OIS — Overnight Index Swap [M4,M5]

Layer 1 — Plain English

An interest rate swap where the floating leg references the compounded overnight risk-free rate (ESTR for EUR, SOFR for USD, SONIA for GBP). OIS rates are the market consensus risk-free rate and are used to discount collateralised derivative cash flows.

Layer 2 — Formula / Structure

$$\text{OIS swap value} = \text{PV(fixed leg at OIS rate)} - \text{PV(compounded daily floating leg over the period)}$$

Layer 3 — Practitioner Usage

Post-LIBOR, OIS rates are the primary market rate for discounting collateralised trades. For uncollateralised trades, a credit-spread-adjusted rate is used instead — the source of FVA. VM CSA cash collateral earns the OIS rate. The ISDA 2020 IBOR Fallbacks Protocol implemented the transition from LIBOR to OIS-based RFRs globally.

INTERVIEW
ANSWER

"OIS stands for Overnight Index Swap — the floating leg references the compounded overnight risk-free rate. Post-LIBOR, it's SOFR in the US, ESTR in the eurozone, SONIA in the UK. OIS rates are the risk-free rate benchmark. For derivatives: if you have a VM cash CSA, you discount cash flows at OIS and earn OIS on posted cash. For uncollateralised trades, you discount at a higher rate — OIS plus your funding spread — and that difference is the source of FVA. The LIBOR to OIS transition in 2021-23 was a massive operational undertaking requiring ISDA protocol adherence and system changes across the entire derivatives industry."

Output Floor — Output Floor — Basel IV RWA Minimum [M1,M6]

Layer 1 — Plain English

A Basel IV requirement that RWA calculated using internal models (IMM, IRB) must be no lower than 72.5% of the standardised approach RWA. The output floor prevents excessive optimisation of internal models and ensures minimum capital comparability across banks.

Layer 2 — Formula / Structure

Effective RWA = max(Internal Model RWA, 72.5% × SA RWA). Phased in: 50% floor in 2025, rising to 72.5% by 2030.

Layer 3 — Practitioner Usage

For derivative-heavy banks that have used IMM to produce low CCR RWA, the output floor materially increases effective RWA and therefore KVA. Banks already on SA-CCR see smaller direct impact. The floor accelerates the shift to CCP clearing where treatment is more capital-efficient. FRTB and the output floor together represent the two biggest structural capital increases from Basel IV for trading book banks.

INTERVIEW
ANSWER

"The Basel IV output floor says internal model RWA cannot be less than 72.5% of standardised RWA. It's the floor under model optimisation — you can't use clever internal models to reduce capital below a minimum level of comparability. For a bank heavily reliant on IMM for CCR, the floor can increase effective RWA significantly. That flows through to KVA: higher RWA means higher capital cost means higher KVA on every derivative trade. The strategic response is to push more business through CCPs, where the capital treatment is better than bilateral even under the floor."

P

P&L Attribution — P&L Attribution — XVA Profit and Loss Decomposition [M6]

Layer 1 — Plain English

The process of decomposing the daily change in XVA mark-to-model value into its constituent drivers: new trades, market moves (rates, spreads, vol), model recalibrations, and portfolio changes. P&L attribution explains where XVA P&L comes from and validates model behaviour.

Layer 2 — Formula / Structure

$$\text{Daily XVA P\&L} = \text{New Trade P\&L} + \text{Theta (time decay)} + \text{Delta P\&L (rate/spread/vol moves)} + \text{Gamma/Convexity} + \text{Recalibration P\&L} + \text{Other}$$

Layer 3 — Practitioner Usage

P&L attribution is both a management information requirement and a regulatory one — FRTB requires P&L attribution tests for IMA approval. For XVA desks, attribution explains whether P&L came from good hedging decisions (desirable) or model sensitivity to recalibration (potentially concerning). CVA theta — the daily release of CVA reserve as trades age — is a significant positive carry component that attribution makes visible. AAD-computed Greeks enable real-time attribution.

INTERVIEW ANSWER

"P&L attribution breaks down where your daily XVA profit or loss came from. Did it come from rate moves (delta P&L)? From counterparty spread moves (CVA CS01 P&L)? From new trades booked? From time passing (theta carry)? From a SIMM recalibration (MVA model P&L)? Each source has a different management response — theta carry is expected and good, unexplained residual P&L signals a model problem. For FRTB IMA, P&L attribution tests are mandatory — if your risk-theoretical P&L (computed from Greeks) doesn't match your actual P&L closely enough, you lose IMA approval. AAD makes the daily Greek computation tractable and therefore attribution feasible."

PD — PD — Probability of Default [M3]

Layer 1 — Plain English

The likelihood that a counterparty will default on its obligations over a specified time horizon. PD is a key CVA input — higher PD means higher expected default loss and therefore higher CVA.

Layer 2 — Formula / Structure

Hazard rate: $h(t) \approx \text{CDS Spread} / \text{LGD}$. **Survival probability:** $S(t) = \exp(-\int_0^t h(s) ds)$. $PD(0, t) = 1 - S(t)$

Layer 3 — Practitioner Usage

Market-implied PD from CDS spreads is used for CVA (consistent with IFRS 13 exit price framework). Accounting-based PD from internal credit ratings is used for IFRS 9 ECL provisioning. For illiquid counterparties without a traded CDS, proxy PD is derived from CDS indices or comparable names. PD volatility — CDS spread moves — is the primary driver of daily CVA P&L.

INTERVIEW ANSWER

"Probability of Default is the likelihood a counterparty fails to pay. For CVA, we use market-implied PD derived from CDS spreads — essentially the market's assessment of credit risk. The approximation: $\text{CDS spread} \approx \text{PD} \times \text{LGD}$. So if the 5-year CDS is 200 bps and LGD is 60%, annual hazard rate is roughly 3.3% and 5-year cumulative PD is about 15%. When CDS spreads widen, PD goes up, CVA goes up, CVA desk books a loss. For counterparties without a traded CDS, you use proxy spreads from comparable rated peers or sector indices — introduce model error but necessary for pricing."

PFE — PFE — Potential Future Exposure [M3,Risk]

Layer 1 — Plain English

The maximum credit exposure a bank could have to a counterparty at a future date, estimated at a specified confidence level (typically 95th or 99th percentile). PFE is the forward-looking component of credit exposure — what the exposure might grow to under adverse market conditions.

Layer 2 — Formula / Structure

PFE (SA-CCR) = multiplier × Aggregate Add-on. **PFE (internal model, 95th pctile)** = 95th percentile of simulated exposure distribution at horizon t

Layer 3 — Practitioner Usage

SA-CCR PFE is used for regulatory capital. Internal model PFE (at 95th percentile) is used for risk limit monitoring (OBSI). The two measures can differ significantly — SA-CCR is conservative and standardised; internal PFE captures specific portfolio characteristics. Higher volatility and longer tenor increase PFE. VM dramatically reduces PFE by limiting the accumulation of unrealised gains.

INTERVIEW ANSWER

"PFE is the potential future exposure — how large your credit exposure might get under adverse market conditions at a future date, at a given confidence level. For limit monitoring you typically use the 95th percentile of simulated exposure over the trade life. For regulatory capital (SA-CCR) you use the standardised add-on formula. The key distinction from MtM: MtM is what you'd lose today; PFE is what you might lose in the future if markets move against the counterparty. OBSI = MtM + PFE + Settlement Risk — PFE is typically the largest component for long-dated uncollateralised trades."

R

RC — RC — Replacement Cost (SA-CCR) [M3]

Layer 1 — Plain English

The current cost of replacing all transactions in a netting set if the counterparty defaulted today — the net present value of the netting set, net of any collateral already held. RC is the "today" component of SA-CCR EAD.

Layer 2 — Formula / Structure

$$RC = \max(V - C, 0) \quad \text{where } V = \text{net MtM of netting set, } C = \text{net eligible collateral held. } EAD = 1.4 \times (RC + PFE)$$

Layer 3 — Practitioner Usage

RC is zero for a fully collateralised portfolio where VM perfectly offsets MtM (zero threshold, zero MTA). VM reduces RC directly. For uncollateralised counterparties RC equals the full positive MtM. RC is the immediate loss; PFE captures future growth of that loss.

INTERVIEW
ANSWER

"Replacement cost is what you'd lose if the counterparty defaulted right now — the current net MtM of all trades, minus any collateral you hold. For a fully VM-collateralised portfolio with zero threshold and zero MTA, RC is near zero because the VM has settled all the accumulated MtM. For an uncollateralised counterparty, RC equals the full in-the-money MtM. RC is the first component of SA-CCR EAD: $EAD = 1.4 \times (RC + PFE)$. RC captures the current snapshot; PFE captures the potential future growth."

Rehypothecation — Rehypothecation — Collateral Reuse [M5]

Layer 1 — Plain English

The practice of reusing collateral received from one counterparty to meet collateral obligations to another. Permitted for VM under most CSAs; explicitly prohibited for UMR-segregated IM.

Layer 3 — Practitioner Usage

Rehypothecation reduces the receiver's funding cost — received bonds can be repo'd out rather than purchasing from market. But it creates systemic chains: if Party A's bonds are rehypothecated by B to C, and C to D, a single failure can cascade. FSB frameworks and SFTR reporting were designed to make these chains visible. Post-UMR, IM cannot be rehypothecated — segregated IM is the posting party's asset even in the receiver's insolvency.

INTERVIEW
ANSWER

"Rehypothecation is when the party that received your collateral reuses it for their own purposes — they repo it out, post it elsewhere, use it as IM. For VM this is typically permitted and it's a significant funding benefit for large dealers. For UMR-segregated IM, it's prohibited — those bonds stay in a third-party custodial account, the receiver can't touch them. The systemic risk point: rehypothecation chains create interconnections across institutions. If A posts to B, B re-posts to C, C re-posts to D, a default at any point stresses the whole chain. That's part of what SFTR reporting — securities financing transactions regulation — is designed to make visible to regulators."

Rho — Rho — Interest Rate Sensitivity [M0]

Layer 1 — Plain English

The sensitivity of a derivative's value to a change in the risk-free interest rate. For options, rho is relatively small compared to delta and vega. For long-dated interest rate derivatives, rho is the dominant sensitivity. Also used to describe the correlation parameter in credit portfolio models.

Layer 2 — Formula / Structure

$\text{Rho} = dV/dr$. For European call: $\text{Rho} = K \times T \times e^{(-rT)} \times N(d_2)$. For an IRS: $\text{Rho} \approx \text{DV01}$ (rate sensitivity dominates).

Layer 3 — Practitioner Usage

Rho matters for long-dated trades where discounting effects are significant. In the XVA context, rho refers both to the interest rate sensitivity of XVA values (discounting changes as rates change) and to the correlation parameter in CVA models (correlation between counterparty PD and market risk factors — the wrong-way risk parameter).

INTERVIEW ANSWER

"Rho measures sensitivity to the risk-free interest rate. For short-dated options it's usually the smallest Greek. For long-dated trades — 10Y+ IRS, long-dated FX forwards — discounting effects make rho significant. In interviews: distinguish two uses of rho. First, the standard Greek: how much does the option price change per 1% move in rates. Second, in credit risk modelling: rho is the asset correlation parameter in the Basel IRB formula and the wrong-way risk correlation in CVA models. Context determines which meaning is intended."

RWA — RWA — Risk-Weighted Assets [M1,M6]

Layer 1 — Plain English

The total of a bank's assets and off-balance sheet exposures weighted by their credit risk, used as the denominator for all regulatory capital ratio calculations. Higher-risk assets carry higher risk weights and therefore consume more regulatory capital.

Layer 2 — Formula / Structure

$\text{CET1 Ratio} = \text{CET1 Capital} / \text{RWA}$ | For derivatives: $\text{RWA} = \text{EAD} \times \text{Risk Weight}$ (BBB = 100%, AA = 20%, AAA = 20%, CCP = 2%). $\text{KVA} = h \times \int \text{RWA}(t) \times \text{CET1}_{\text{req}} \times df(t) dt$

Layer 3 — Practitioner Usage

RWA is the central driver of KVA — every unit of RWA generates a capital cost of $\text{CET1}_{\text{req}} \times \text{hurdle rate}$ per year. RWA reduction through clearing, netting, compression, and collateral directly reduces KVA. Basel

IV output floor constrains how low internal model RWA can go (72.5% of SA RWA). For a BBB corporate, risk weight is 100% under Basel SA — EAD equals RWA.

INTERVIEW ANSWER

"RWA is risk-weighted assets — the denominator of all capital ratios. It weights each exposure by its riskiness: a AAA government bond gets 0%, a BBB corporate derivative gets 100%, equity gets 100-150%. The higher the RWA, the more capital you need to hold. For KVA: every EUR of RWA costs the bank $CET1_{req} \times hurdle$ per year. At 8.5% CET1 and 12% hurdle, that's 1.02% per year per EUR of RWA. Reduce RWA by clearing or netting, and KVA falls proportionally. That's why balance sheet optimisation — clearing, compression — is a direct KVA management tool."

RWR — RWR — Right-Way Risk [M3]

Layer 1 — Plain English

The favourable scenario where counterparty default probability decreases as the trade exposure increases — the counterparty is least likely to default precisely when the bank's exposure is highest. RWR reduces CVA below the independent baseline.

Layer 3 — Practitioner Usage

Right-way risk reduces CVA and creates pricing opportunities. A gold mining company buying gold price protection: as gold prices rise, the protection becomes more in-the-money (increasing exposure) but the gold miner's creditworthiness also improves (lower PD). The net effect is RWR — reduced CVA. Correctly modelling RWR produces lower CVA and more competitive pricing vs a desk that ignores correlation.

INTERVIEW ANSWER

"Right-way risk is the benign version of wrong-way risk — exposure increases when default probability decreases. The gold miner example: they buy you gold price protection. If gold rallies, your exposure grows but so does their creditworthiness (they're making money). The two move opposite to each other — that's right-way risk and it reduces CVA below the independent benchmark. In practice, identifying right-way risk in your portfolio is a source of competitive pricing advantage. You can offer tighter levels than a competitor who ignores the correlation. Modelling it requires joint simulation of market risk factors and credit risk."

S

SA-CCR — SA-CCR — Standardised Approach for Counterparty Credit Risk [M3]

Layer 1 — Plain English

The Basel III standardised methodology for computing EAD for OTC derivatives and securities financing transactions. SA-CCR replaced the old Current Exposure Method (CEM), introducing more risk-sensitive treatment of netting, collateral, and asset class differences.

Layer 2 — Formula / Structure

$$\text{EAD} = \alpha \times (\text{RC} + \text{PFE}) \quad | \quad \alpha = 1.4 \quad | \quad \text{RC} = \max(\text{V}-\text{C}, 0) \quad | \quad \text{PFE} = \text{multiplier} \times \text{Aggregate Add-on} \quad | \quad \text{Agg Add-on} = \sum \text{asset class add-ons with partial netting}$$

Layer 3 — Practitioner Usage

Key improvements over CEM: (1) netting benefit recognised in PFE (not just RC); (2) VM reduces RC directly; (3) IM reduces PFE via the multiplier; (4) more granular asset class supervisory factors. SA-CCR is the capital engine for KVA — $E[K(t)]$ is derived from projected SA-CCR EAD over the trade life. Banks on SA-CCR generally have higher RWA than those with IMM approval — but IMM is subject to the Basel IV output floor.

INTERVIEW ANSWER

"SA-CCR is the standardised formula for counterparty credit risk EAD — it's alpha of 1.4 times the sum of replacement cost and potential future exposure. The key improvements over the old CEM: SA-CCR recognises netting in the PFE calculation (CEM didn't), VM reduces RC directly, and IM reduces PFE through the multiplier. For KVA: SA-CCR EAD drives RWA, which drives the capital cost. You project SA-CCR EAD forward over the trade life to get the $E[K(t)]$ profile, multiply by CET1_req to get expected capital, multiply by the hurdle rate and discount back. That's KVA."

SCB — SCB — Stress Capital Buffer [M1,M6]

Layer 1 — Plain English

A US Federal Reserve dynamic capital buffer introduced in 2020 for large US bank holding companies, replacing the fixed 2.5% Capital Conservation Buffer with a buffer derived from each bank's annual DFAST stress test results. The SCB minimum is 2.5%.

Layer 2 — Formula / Structure

$$\text{SCB} = \max(2.5\%, \text{peak-to-trough CET1 decline in the severely adverse DFAST scenario})$$

$$\text{Total CET1}_{\text{req}} \text{ for a US G-SIB} = 4.5\% + \text{SCB} + \text{G-SIB surcharge}$$

Layer 3 — Practitioner Usage

The SCB creates a direct link between macro stress test results and individual bank capital requirements — and therefore KVA. A bank with a high SCB (driven by stress test sensitivity) pays higher KVA on every trade. European banks face the SREP Pillar 2 add-on instead. The SCB introduces year-to-year variability in KVA reserves that requires active management by the XVA desk.

INTERVIEW ANSWER

"The SCB replaces the fixed 2.5% conservation buffer for large US banks with a dynamic buffer derived from the annual DFAST stress test. The worse your stress test performance, the larger your SCB. It creates a direct link between macro scenarios and trade-level capital costs — your KVA literally changes each year based on stress test outcomes. The minimum is 2.5% so you're never better than the fixed CCB equivalent, but you could be much worse. European banks face the SREP Pillar 2 add-on instead — same concept, different mechanics. For KVA desks at US G-SIBs, the annual DFAST results are a major calendar event."

Settlement Risk — Settlement Risk — Herstatt Risk [Risk]

Layer 1 — Plain English

The risk that a counterparty fails to deliver the security or cash on the settlement date after the reporting entity has already made its own delivery. Named after Bankhaus Herstatt, which failed in 1974 mid-settlement, leaving counterparties with USD payments made but DEM not received.

Layer 2 — Formula / Structure

$$\text{Settlement Risk Exposure} = \text{Full value of the outstanding leg awaiting delivery, during the settlement window (T+1 to T+2)}$$

Layer 3 — Practitioner Usage

Settlement risk is highest on the settlement date of large transactions and drops to zero once both legs are confirmed settled. CLS (Continuous Linked Settlement) eliminated most FX settlement risk for its

member currencies by synchronising payment and receipt. For bonds and OTC derivatives, settlement risk persists during the T+1 or T+2 window. Settlement risk forms the third component of OBSI.

INTERVIEW ANSWER

"Settlement risk, also called Herstatt risk after the 1974 German bank failure, is the risk that you've paid but haven't received. In the Herstatt case, German banks had paid DEM to Herstatt but the USD payments hadn't yet been made when US regulators closed the bank — leaving US counterparties short. CLS Bank was created specifically to solve FX settlement risk by synchronising both legs. For bonds and OTC derivatives, settlement risk still exists during the T+1/T+2 window. It's the 'in transit' component of OBSI — typically small but spikes on large settlement dates."

SFTR — SFTR — Securities Financing Transactions Regulation [M5]

Layer 1 — Plain English

EU/UK regulation requiring daily reporting of all securities financing transactions (repo, reverse repo, securities lending, margin lending) to a trade repository. SFTR makes the rehypothecation chain and collateral reuse visible to regulators, addressing the information asymmetry that amplified the 2008 crisis.

Layer 3 — Practitioner Usage

SFTR reports over 150 data fields per transaction including collateral details, haircuts, reuse information, and counterparty identifiers. It applies to EU/UK-regulated counterparties including banks, asset managers, and insurance companies. SFTR reporting gives regulators the data to map the interconnections created by rehypothecation chains — chains that were invisible before 2008. The FSB uses SFTR data to monitor systemic risk from securities financing.

INTERVIEW ANSWER

"SFTR requires daily reporting of all securities financing transactions — repos, reverse repos, securities lending, margin lending — to an approved trade repository. The regulatory rationale: before 2008, regulators had no visibility into the rehypothecation chains that created systemic interconnections. SFTR provides 150+ fields per transaction including collateral details and reuse information. For derivatives practitioners: SFTR interacts with the collateral transformation business. If you're running a collateral upgrade trade (IG bonds → government bonds), that's an SFT and must be reported. SFTR reporting obligations apply to EU/UK entities; US equivalents exist under SEC Rule 10c-1 for securities lending."

SIMM — SIMM — Standard Initial Margin Model [M4]

Layer 1 — Plain English

The ISDA industry-standard model for calculating bilateral Initial Margin under UMR. SIMM computes IM based on trade sensitivities (greeks) across six risk classes, applying prescribed risk weights and correlation parameters. SIMM v2.8+2506 (December 2025) is the current version.

Layer 2 — Formula / Structure

$$\text{SIMM IM} = \sqrt{[\Sigma_i (WS_i^2) + \Sigma_{i \neq j} (\rho_{ij} \times WS_i \times WS_j)] \text{ across 6 risk classes: IR, CR-Q, CR-NQ, EQ, FX, CM}}$$

Layer 3 — Practitioner Usage

SIMM is more capital-efficient than the regulatory schedule (grid) approach because it recognises diversification across risk classes. Bilateral SIMM reconciliation is performed via AcadiaFX to resolve disputes. Semi-annual recalibrations directly affect MVA. New trades that net against existing positions reduce aggregate SIMM IM.

INTERVIEW ANSWER

"SIMM is the industry-standard model for computing bilateral Initial Margin under UMR. It works by taking trade sensitivities — the greeks — and applying ISDA-prescribed risk weights and correlations across six risk classes. The result is an IM amount that reflects the portfolio's actual risk profile and recognises diversification — unlike the simpler grid approach which ignores it. For MVA, the SIMM IM profile over the trade life is the key driver. Semi-annual recalibrations are a calendar event for XVA desks — if risk weights increase, MVA reserves increase, P&L takes a hit. Bilateral reconciliation via AcadiaFX ensures both parties are working from consistent SIMM results."

SIMM Recalibration — SIMM Recalibration — Semi-Annual Risk Weight Updates ^[M4]

Layer 1 — Plain English

The semi-annual process by which ISDA recalibrates the risk weights and correlation parameters in the Standard Initial Margin Model (SIMM) to reflect current market conditions. Recalibrations directly affect IM requirements and therefore MVA for all in-scope bilateral relationships.

Layer 2 — Formula / Structure

$$\text{IM change from recalibration} = \Delta \text{SIMM_IM} = \text{New Risk Weights} \times \text{Sensitivities} - \text{Old Risk Weights} \times \text{Sensitivities}. \text{ MVA impact} = -\int \Delta \text{SIMM_IM}(t) \times s_{\text{fund}} \times df(t) dt$$

Layer 3 — Practitioner Usage

SIMM recalibrations in December and June each year are significant calendar events for XVA desks. A recalibration that increases IR delta risk weights (e.g. following a period of elevated rate volatility) increases IM for all IRS books simultaneously, triggering a system-wide MVA increase and P&L impact. Banks must implement new SIMM versions by the ISDA-specified effective date — operational readiness for semi-annual updates is a regulatory expectation.

INTERVIEW ANSWER

"SIMM recalibration happens twice a year — December and June — when ISDA updates the risk weights and correlations in the model to reflect recent market conditions. For XVA desks this is a major event: if IR risk weights increase (say, because rates have been volatile), all IRS IM goes up simultaneously, increasing MVA reserves and booking a P&L loss. The reverse is also true — lower risk weights after a calm period reduce IM and release MVA P&L. Operationally, banks must implement the new SIMM version by the effective date, which requires model governance sign-off, system updates, and bilateral reconciliation with counterparties via AcadiaFX. It's a well-flagged calendar event but the P&L impact can still be significant."

T

Tenor Limit — Tenor Limit — Maximum Maturity Limit [Risk]

Layer 1 — Plain English

A credit limit that constrains the maximum remaining maturity of transactions permitted with a counterparty. Tenor limits prevent the accumulation of very long-dated exposure to counterparties where credit quality may deteriorate significantly over the extended horizon.

Layer 3 — Practitioner Usage

Tenor limits interact with PFE limits: a 5Y tenor limit with a large OBSI limit means short-dated trades are unrestricted but long-dated trades require senior credit approval. KVA grows with tenor — so tenor limits also indirectly constrain capital consumption. For counterparties with weaker credit quality, shorter tenor limits reduce the PFE uncertainty inherent in long-dated exposures.

INTERVIEW
ANSWER

"A tenor limit caps the maximum maturity of trades you can do with a counterparty. The rationale: credit quality can change significantly over 10 or 20 years, and long-dated PFE is inherently uncertain. A BBB counterparty today might be high yield in 5 years. Tenor limits are often calibrated by credit rating — investment grade counterparties might have a 10-year limit, high yield only 2 years. KVA also grows with tenor, so tenor limits have a capital efficiency dimension beyond just credit risk control."

Theta — Theta — Time Decay [M0]

Layer 1 — Plain English

The rate of change of a derivative's value with respect to the passage of time, holding all other factors constant. For long option positions, theta is negative — the option loses value daily as it approaches expiry. Often expressed as daily P&L impact from time passing.

Layer 2 — Formula / Structure

Theta = dV/dt (expressed as: change in value per 1 day of time passing). For ATM European options: $\text{Theta} \approx -S \times \sigma \times N'(d1) / (2\sqrt{T})$

Layer 3 — Practitioner Usage

Theta and gamma have an inverse relationship for long options: long gamma (convexity benefit) is paid for via negative theta (daily decay). XVA has an implicit theta: as a trade ages toward maturity, EPE declines and XVA reserves release — generating positive carry P&L. KVA theta is the daily release of the capital cost reserve as $E[K(t)]$ declines with time.

INTERVIEW
ANSWER

"Theta is daily time decay — how much an option loses in value from one day passing. For long options it's negative: you pay theta to own the convexity. The theta-gamma tradeoff is central to options trading: long gamma means you profit from large moves, but you pay for it via daily theta erosion. For XVA: there's an implicit positive theta as the trade ages and reserves release. KVA theta — the daily release of capital cost reserves as the trade gets closer to maturity — is a real P&L line item on XVA books. It's positive carry that accrues daily and is realised as the trade matures."

Threshold (CSA) — Threshold — CSA Credit Exposure Threshold [M4,M5]

Layer 1 — Plain English

The amount of exposure that a counterparty may have before any collateral is required under a CSA. A zero threshold means full collateralisation from the first dollar of exposure; a positive threshold creates a permanently uncollateralised band up to that amount.

Layer 2 — Formula / Structure

**VM call is triggered when: $\text{Net MtM} - \text{Collateral held} > \text{Threshold}$.
Residual uncollateralised exposure $\approx \text{Threshold (persistently)} + \text{MTA band (randomly)}$.**

Layer 3 — Practitioner Usage

Thresholds are often credit-rating-linked — higher for investment grade counterparties, lower or zero for high yield. A rating downgrade can trigger an automatic threshold reduction, which in turn triggers a large VM call — a credit event waterfall. For CVA and FVA modelling, the threshold defines the permanent floor of uncollateralised EPE that persists even for CSA counterparties. BCBS-IOSCO UMR rules require zero threshold for IM (though VM thresholds can still be non-zero under some regulatory frameworks).

INTERVIEW ANSWER

"The threshold is the amount of exposure allowed before VM is required. Zero threshold: full collateralisation from day one — CVA and FVA approach zero. EUR 5M threshold: up to EUR 5M of MtM can accumulate before the counterparty has to post collateral — that EUR 5M is permanently uncollateralised and carries CVA and FVA. Thresholds are often rating-linked — a BBB counterparty might have a EUR 5M threshold that drops to EUR 1M if they're downgraded to BB. That automatic threshold reduction triggers a large VM call when the downgrade occurs — a significant operational and liquidity event. Always model the threshold explicitly when computing XVA for CSA counterparties."

U

UMR — UMR — Uncleared Margin Rules [M4,M5]

Layer 1 — Plain English

The BCBS-IOSCO framework requiring mandatory exchange of Variation Margin and Initial Margin for non-cleared OTC derivatives between in-scope counterparties. Phased in across six tranches from 2016 to 2022 based on each entity's Average Aggregate Notional Amount (AANA).

Layer 2 — Formula / Structure

IM threshold: Phase 1 (AANA $\geq$ EUR 3T, 2016) → Phase 6 (AANA $\geq$ EUR 8B, 2022). All phases now active.

Layer 3 — Practitioner Usage

UMR Phase 5 and 6 (2021–2022) dramatically expanded IM requirements to mid-tier buy-side firms. UMR prohibits rehypothecation of segregated IM — the locked collateral cost is the source of MVA. The ISDA 2016 VM Protocol was the largest regulatory protocol adoption in ISDA history. Post-UMR, bilateral IM infrastructure (SIMM, custodians, AcadiaFX) became standard industry infrastructure.

INTERVIEW ANSWER

"UMR is the regulatory requirement for bilateral Initial Margin on non-cleared OTC derivatives. It was phased in from 2016 to 2022 — large dealers first, progressively capturing smaller buy-side firms. The key features: IM must be calculated using SIMM or the regulatory grid, must be segregated at a third-party custodian, and cannot be rehypothecated. That last point is the source of MVA — the locked collateral has a funding cost that must be priced into the trade. Phase 5 and 6 in 2021-22 brought in hundreds of new entities and triggered massive industry investment in SIMM infrastructure and custodial arrangements."

Utilisation — Utilisation — Limit Utilisation Percentage [Risk]

Layer 1 — Plain English

The percentage of a credit or market risk limit currently consumed by outstanding exposure. Utilisation = (Current Exposure / Limit) $\times$ 100. Real-time utilisation monitoring against approved limits is the primary function of credit control.

Layer 2 — Formula / Structure

**Utilisation % = (MtM + PFE + Settlement Risk) / OBSI Limit $\times$ 100.
Headroom = Limit $\times$ (1 - Utilisation/100)**

Layer 3 — Practitioner Usage

Utilisation thresholds typically trigger escalating alerts: 80% (first alert — credit officer notified), 90% (second alert — business head notified), 95% (trading desk alert — new business at risk), 100%+ (breach — escalation required, new business frozen).

INTERVIEW ANSWER

"Utilisation is how much of the limit you're using — current exposure divided by the limit, expressed as a percentage. 100% means you're at the limit; above 100% is a breach. Alert thresholds are typically set at 80%, 90%, and 95% — each level triggers a different escalation. Credit controllers watch utilisation in real-time across the whole book. The proactive management principle: ideally you never reach 100% because you've been managing utilisation down as it approaches the limit, rather than reacting to a breach after it happens."



VaR — VaR — Value at Risk [M2]

Layer 1 — Plain English

A statistical measure of the maximum potential loss over a specified time horizon at a given confidence level. A 1-day 99% VaR of EUR 10M means there is a 1% chance of losing more than EUR 10M on any given day. VaR has been the primary market risk measure for over 30 years.

Layer 2 — Formula / Structure

$VaR(\alpha, T) = \text{Quantile}(\alpha) \text{ of the loss distribution over horizon } T.$
For normal distribution: $VaR = \mu + z_{\alpha} \times \sigma \times \sqrt{T}$. 1-day 99% VaR $\approx 2.33 \times \text{daily P\&L standard deviation (if zero mean)}$

Layer 3 — Practitioner Usage

VaR was the Basel II and Basel 2.5 primary market risk capital measure. FRTB replaced VaR at 99% with ES at 97.5% as the capital metric. VaR remains widely used for internal risk management, limit setting, and P&L attribution despite its key weakness: it says nothing about the severity of losses beyond the confidence threshold (which ES addresses). Backtesting VaR at 99% requires counting days where actual losses exceed VaR — no more than 4 in 250 days for the green zone.

INTERVIEW
ANSWER

"VaR is the loss you won't exceed X% of the time over a given horizon. 1-day 99% VaR of EUR 10M: on 99 days out of 100, you lose less than EUR 10M. On the 100th day, you might lose much more — VaR says nothing about how much. That's the critical weakness ES addresses. For regulatory purposes, FRTB moved from VaR to ES precisely for this reason. VaR is still used for internal limits and P&L attribution. Backtesting: count exceptions over 250 days — 0-4 is green zone, 5+ starts attracting capital multiplier increases. The traffic light system is a simple but effective model validation tool."

Vega — Vega — Volatility Sensitivity [M0]

Layer 1 — Plain English

The sensitivity of a derivative's value to a change in implied volatility. Vega is positive for long options (options gain value as vol rises) and zero for linear instruments (swaps, forwards). Vega is one of the most commercially significant Greeks for options dealers.

Layer 2 — Formula / Structure

$\text{Vega} = dV/d\sigma$ | For European options: $\text{Vega} = S \times \sqrt{T} \times N'(d1)$ (per 1% absolute vol move). ATM options have highest vega.

Layer 3 — Practitioner Usage

Vega is a key risk for XVA desks — higher implied vol increases EPE and therefore CVA, FVA, and KVA. ColVA has a vega ($d\text{ColVA}/d\text{SpreadVol}$) — rising cross-currency basis vol increases the CTD spread option value. KVA vega ($d\text{KVA}/d\text{Vol}$) requires AAD for production-scale computation. FRTB SA capital includes a vega margin component for options books.

INTERVIEW
ANSWER

"Vega measures how much your option position changes value for a 1% move in implied volatility. Long options have positive vega — you benefit when vol rises. The practical importance: if you're long a straddle with EUR 500,000 of vega and vol rises 2%, you make EUR 1M. For XVA: higher vol increases EPE (future market moves are larger), which increases CVA. The ColVA connection: the CTD spread option in a multi-currency CSA has a vega — higher cross-currency basis vol makes the option more valuable. Vega across a large derivatives book can be an enormous risk that requires active hedging via swaptions and variance swaps."

VM — VM — Variation Margin [M4,M5]

Layer 1 — Plain English

The daily cash or securities settlement of accumulated mark-to-market changes in a derivative portfolio under a CSA. VM eliminates the build-up of unrealised gains and losses, dramatically reducing CVA and FVA to near zero for fully collateralised books.

Layer 2 — Formula / Structure

VM call = max(Net MtM change since last call - MTA, 0) subject to Threshold and Rounding. Cash VM earns OIS rate (ESTR/SOFR/SONIA) per CSA terms.

Layer 3 — Practitioner Usage

VM can be rehypothecated (unlike UMR-segregated IM). For bank-to-bank trades under a zero-threshold daily VM cash CSA, CVA and FVA approach zero. The ISDA 2016 VM Protocol aligned bilateral VM terms with regulatory requirements across hundreds of thousands of trading relationships in one coordinated adherence process.

INTERVIEW ANSWER

"Variation Margin is daily cash settlement of accumulated MtM — it keeps the trade close to zero MtM every day rather than letting gains and losses accumulate. The XVA impact is enormous: with a zero-threshold daily VM cash CSA, CVA and FVA go to near zero because there's no accumulated exposure for default risk or funding cost to attach to. That's why the ISDA 2016 VM Protocol adherence was such a major industry event — it brought hundreds of thousands of relationships into daily VM compliance simultaneously. VM is also rehypothecable, unlike IM, which is why it's cheaper to fund than segregated Initial Margin."

W

WWR — WWR — Wrong-Way Risk [M3]**Layer 1 — Plain English**

The risk that counterparty default probability is positively correlated with exposure at the time of default — the counterparty is most likely to default precisely when the trade is most in-the-money for the bank. WWR amplifies CVA beyond the independent $EPE \times PD$ baseline.

Layer 2 — Formula / Structure

$$\text{CVA_WWR} > \text{CVA_independent} \text{ when } \text{Corr}(\text{EPE}, \text{PD}) > 0. \text{ WWR multiplier} \approx \frac{1 + \rho \times \sigma_{\text{EPE}} \times \sigma_{\text{PD}}}{(\text{EPE} \times \text{PD})} \text{ (simplified)}$$

Layer 3 — Practitioner Usage

Classic WWR examples: energy producer selling commodity price protection (producer most likely to default when commodity prices are low — exactly when protection is most in-the-money); sovereign CDS buyer referencing the sovereign's own bonds as collateral. ISDA WWR guidelines provide qualitative and quantitative frameworks. SA-CCR addresses specific WWR patterns via supervisory factors. Wrong-way CSA collateral (counterparty posts bonds of an entity correlated with their own default) amplifies WWR.

INTERVIEW ANSWER

"Wrong-way risk is the dangerous correlation where the counterparty is most likely to default exactly when your exposure is highest. The textbook example: an energy producer has sold you oil price protection. When oil prices crash, the protection is most valuable to you — but the producer is most likely to be in financial distress. So your exposure peaks when default probability peaks. That correlation amplifies CVA far beyond the simple $\text{EPE} \times \text{PD}$ calculation. The right-way equivalent — where high exposure correlates with low PD — reduces CVA. Wrong-way risk is why CVA can't just be a simple formula and why counterparty selection and trade structuring matter."

X

XVA — XVA — X-Valuation Adjustment [M3,M4,M5,M6]

Layer 1 — Plain English

The collective term for all valuation adjustments applied to derivative prices beyond risk-free discounted cash flows. The X is a wildcard representing any specific adjustment. The XVA stack — CVA, DVA, FVA, MVA, ColVA, KVA — captures the full economic cost of entering and maintaining a derivative position.

☉ XVA Waterfall — Complete Stack

Risk-free price (OIS discounted)

CVA	Expected cost of counterparty default (-) cost to dealer	[M3]
DVA	Own default benefit (mirror of CVA) (+) benefit to dealer (controversial)	[M3]
FVA	Funding cost of uncollateralised exposure (-) cost; zero for fully VM-collateralised	[M4]
MVA	Funding cost of posting Initial Margin (-) cost; zero if no IM required	[M4]
ColVA	Value of CTD optionality in multi-CCY CSA (+) for poster; (-) for receiver; 0 for single-CCY	[M6]
KVA	Cost of regulatory capital over trade life (-) always a cost; K = Kapital (German)	[M6]
ALL-IN PRICE = Risk-free - CVA + DVA - FVA - MVA + ColVA - KVA		

Layer 2 — Formula / Structure

Total XVA = CVA - DVA + FVA + MVA - ColVA (poster benefit) + KVA.
All components integrated over trade life and discounted at OIS.

Layer 3 — Practitioner Usage

XVA emerged post-2008 as banks recognised that derivative prices had to reflect the full economic cost of the position. The XVA waterfall covered in this series spans Modules 3–6. XVA desks at G-SIBs manage aggregate reserves in the billions. Under stress, XVA components can fail simultaneously — the admissibility framework (enforceable × fundable × capital-feasible) provides the correct lens for regime-shift scenarios.

INTERVIEW ANSWER

"XVA is the umbrella term for all the adjustments that go on top of the risk-free price of a derivative: CVA for counterparty default, DVA for your own default, FVA for funding cost, MVA for IM funding cost, ColVA for collateral optionality, KVA for capital cost. Before 2008 most of these were either ignored or bundled into the bid-offer spread. Post-crisis, banks had to explicitly price and reserve for each component — creating the XVA desk as a distinct business function. The total XVA on a trade determines whether it's actually profitable after accounting for all the real economic costs. A trade that looks profitable at the risk-free level might be loss-making once XVA is included."

XVA Desk — XVA Desk — Centralised Valuation Adjustment Function ^[M6]

Layer 1 — Plain English

The centralised function at a major derivatives dealer responsible for pricing, reserving, hedging, and P&L management of all XVA components across the full derivative book. The XVA desk acts as the internal market-maker for credit, funding, margin, and capital costs.

Layer 3 — Practitioner Usage

XVA desks at G-SIBs manage aggregate reserves that can run into billions of dollars. They employ quants (model development), technologists (AAD, GPU computing, real-time calculation), traders (hedging CVA/FVA/CoVA), and risk managers (P&L attribution, limits). The desk structure varies: G-SIBs typically centralise all XVA in one desk; regional banks may distribute ownership to originating desks with XVA oversight. CVA hedgeability, FVA accounting treatment, and KVA admissibility under stress are the three strategic debates that define a sophisticated XVA function.

INTERVIEW ANSWER

"The XVA desk is the centralised hub that prices, reserves for, and hedges all the XVA adjustments across the whole derivative book. It's a hybrid function — part trading desk (actively hedging CVA credit delta and CoVA cross-currency basis), part finance function (managing reserves), part technology function (maintaining real-time AAD-powered Greeks infrastructure). The career path is highly valued because it spans quant finance, technology, risk management, and regulatory capital simultaneously. The key interview insight: an XVA desk at a G-SIB is not a cost centre — it actively generates P&L through the carry of XVA reserves, hedging gains, and intelligent balance sheet management."

Credits — Full Series

Curated by Sarthaka Nanda (with Anthropic Claude)

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M0	Greeks & Sensitivities
M1	Basel Accord Evolution — Basel I through Basel IV
M2	FRTB — Fundamental Review of the Trading Book
M3	CVA, DVA & Counterparty Credit Risk
M4	FVA & MVA — Funding & Margin Valuation Adjustments
M5	ISDA, Netting & Collateral
M6	KVA, CoIVA & Capital Costs — Completing the XVA Waterfall
MD	Practitioner's Dictionary ✓ THIS VOLUME